

Pairing, Stripes, and Competing Order in Coupled Hubbard Ladders

Introduction, Background, and Current Results

Ladder MPS+MF project
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Abstract

Weakly coupled two-leg Hubbard ladders provide a setting in which the formation of singlet pairs can be studied separately from the collective order selected by coupling correlated subsystems. This draft develops the motivation for a multichannel matrix-product-state plus mean-field study of superconducting, charge, and magnetic order, with particular attention to the relationship between stripe-centered pairing proposals and the broader picture of intertwined orders. It also collects the project's current local evidence. The isolated repulsive ladder exhibits a finite-system spin gap and substantial pairing correlations, whereas the latest square-array trajectories at zero nearest-neighbor interaction develop strong stripe textures. At an attractive interaction, two contrasting initial states instead approach a common paired state. The completed unfrustrated-cubic grid instead reaches stripes at all nine tested coordinates. Finer square cuts retain distinct textures near the boundary, with a small change in energy slope along the interaction cut but no resolved first-order singularity. All four repulsive- V square starts lose pairing, including both intertwined seeds. Both trellis seeds become paired in a skew one-ladder cell but develop stripes in a rectangular two-ladder cell, where large alternating relaxation persists. These observations motivate a distinction between local pairing tendencies and the order selected in an array. The numerical results remain finite-length, finite-bond-dimension evidence, with explicit limits on self-consistency, response completeness, and thermodynamic interpretation.

Draft scope. Sections 1–2 are draft manuscript prose. Section 3 is a dated collection of results and interpretation, including unresolved points that must remain visible during manuscript development. The evidence cutoff is the synchronized analyses through 19 September 2026; the literature discussion incorporates the stripe and ladder-material discussions through 18 September. Appendix A identifies the local sources. No new many-body calculation is reported here.

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1 Introduction

The relationship between superconductivity, antiferromagnetism, and spatial charge organization is a central question in doped Mott insulators. The Hubbard model provides a minimal language for studying how kinetic energy and local repulsion generate collective behavior, while its extensions allow the influence of nonlocal interactions and lattice structure to be examined systematically [1–4]. In this setting, strong singlet correlations need not lead directly to superconductivity. Charge and spin order can develop from the same interacting electrons, and the scales associated with pair formation and collective phase coherence can differ. A microscopic account of superconductivity must therefore address the competing ways in which a correlated electronic state can organize.

Stripe order makes this issue particularly concrete. Experiments in lanthanum-based cuprates established charge modulations accompanied by incommensurate magnetic order, with antiferromagnetic domains separated by hole-rich regions [5]. Subsequent measurements found closely related magnetic excitation spectra in stripe-ordered and superconducting materials, motivating theories in which stripe correlations participate in the pairing mechanism [6]. A complementary view regards stripes as a competing organization of a doped Mott insulator whose short-distance singlet physics can also support superconductivity without requiring a particular stripe arrangement [3]. The resulting question concerns both the origin of pairing and the consequences of charge organization for the coherence of pairs.

The distinction between these views is sharpened by separating the existence of stripe correlations from their necessity for superconductivity. In stripe-ordered $\text{La}_{1.875}\text{Ba}_{0.125}\text{CuO}_4$, transport and magnetization measurements reveal superconducting correlations within the planes despite strong suppression of three-dimensional phase coherence [7]. A spatially sign-changing superconducting order parameter can frustrate interplane Josephson coupling and offers a specific explanation of this separation [8]. Such observations permit cooperation at the level of local pairing and competition at the level of collective order. They motivate treating superconducting, charge, and spin degrees of freedom together, as emphasized in the framework of intertwined orders [9, 10].

Modern Hubbard-model calculations reinforce the dependence of this relationship on the microscopic Hamiltonian. Complementary numerical methods find stripe order with small energy differences between wavelengths in a studied regime of the nearest-neighbor square-lattice model [11]. Other calculations find no superconducting ground state in specified regimes of that minimal model [12], while additional hopping terms can support pairing together with spatial charge modulation [13, 14]. Deliberately imposed hopping inhomogeneity can also enhance superconducting correlations on cylinders [15]. These results do not give a universal sign to the influence of stripes. They instead identify a competition that must be resolved at the level of a specified model, geometry, and numerical accuracy.

Two-leg ladders provide useful correlated building blocks for this problem. Their magnetic spectrum differs qualitatively from that of a single chain, and doped repulsive ladders can support a spin gap, bound holes, and slowly decaying singlet-pair correlations [16–18]. The corresponding low-energy description can contain a gapped spin sector and a gapless total-charge mode [19, 20]. High-accuracy numerical studies show that the relative decay of pair and density correlations requires careful control of open boundaries, bond dimension, and fit windows [21, 22]. Thus a ladder can exhibit substantial pairing tendencies while retaining the fluctuations characteristic of an effectively one-dimensional system.

Cuprate ladder materials make the role of transverse geometry an experimental question. Superconductivity was discovered under pressure in $\text{Sr}_{0.4}\text{Ca}_{13.6}\text{Cu}_{24}\text{O}_{41.84}$, a member of the $\text{Sr}_{14-x}\text{Ca}_x\text{Cu}_{24}\text{O}_{41}$ chain-ladder family [23]. The same family exhibits ladder charge order and

pressure-dependent transport, providing a setting in which pairing, charge organization, and interladder coherence can be compared [24–26]. Its ladder sheets have displaced neighbors connected by zigzag paths: a trellis network rather than an array with only aligned transverse bonds. The chain subsystem also acts as a charge reservoir, so varying composition or pressure changes more than the coupling between otherwise fixed ladders. These distinctions motivate a controlled comparison of transverse connectivities before attempting a material-specific phase diagram.

Coupling such ladders introduces a second question: which order develops when the subsystems reinforce one another? Stripe-centered theories provide an explicit route from spin-gap formation to pairing and then to Josephson coherence [27]. A spatially modulated repulsive Hubbard construction demonstrates how nearly decoupled ladders can supply a pairing scale while weak interladder coupling establishes superconducting order [28]. Matrix-product-state plus mean-field methods provide a practical numerical realization of the broader coupled-subsystem strategy [29, 30]. Their multichannel extension shows why charge and superconducting solutions must be retained and compared within a common framework [31].

The interaction between neighboring densities supplies an additional tuning parameter. Calculations on extended Hubbard ladders and cylinders find that attraction can enhance pairing over selected parameter ranges, while strong attraction can also favor charge segregation [32, 33]. Recent cuprate-ladder spectroscopy, interpreted together with DMRG, provides experimental motivation for considering attraction beyond the simple onsite Hubbard interaction [34]. Its implications for a coupled array remain to be determined: stronger binding changes both the properties of the individual ladder and the effective interladder couplings. Pair mobility, susceptibility, and competing order must be considered along with binding.

The present project investigates these issues in arrays of repulsive two-leg extended Hubbard ladders, treating the ladder with an MPS ground-state method and the transverse environment with a multichannel mean-field construction. The principal regime is $U/t = 8$ and density $n = 15/16$ per physical site. Rung hopping, nearest-neighbor interaction, and transverse connectivity are varied while pairing, normal exchange, and spin-resolved density fields remain available. In particular, the trellis comparison tests whether zigzag coupling changes the balance between static stripes and pairing at the same intraladder parameters, using both one- and two-ladder spatial cells. The immediate objective is to establish reproducible ordered solutions and understand how their competition relates to the underlying ladder correlations. The current evidence suggests that intrinsic pairing tendencies can coexist with an array tendency toward static stripes, and that attraction can reverse the observed outcome at a representative square-array point. Establishing the extent and thermodynamic stability of these behaviors requires the numerical controls discussed below.

2 Physical background and connection to the model

2.1 Local singlets, pairing correlations, and coherence

Three observables answer different questions. A spin gap measures the energy cost of the lowest accessible spin excitation at fixed particle number. A pair-binding energy compares even- and odd-particle sectors. A pair correlation function measures how a pair created in one region correlates with a pair removed elsewhere. None of these quantities alone establishes a nonzero superfluid stiffness or a superconducting transition temperature. For example, an isolated number-conserving

ladder has zero anomalous expectation value even when its pair correlations decay algebraically.

For a spin-singlet bond operator,

$$\Delta_{ab} = \frac{c_{a\uparrow}c_{b\downarrow} - c_{a\downarrow}c_{b\uparrow}}{\sqrt{2}}, \quad P_{ab,cd} = \langle \Delta_{ab}^\dagger \Delta_{cd} \rangle, \quad (1)$$

the relative signs between different bond orientations characterize the internal pair structure. Opposite signs on leg and rung bonds are commonly described as *d*-wave-like in a two-leg ladder. The reduced lattice symmetry and discrete transverse momentum should be kept in mind when connecting this terminology to the gap of a two-dimensional cuprate.

An order parameter with a nonzero center-of-mass wavevector is a different property. Schematically,

$$\langle \Delta_b(R) \rangle = f_b [\Psi_0 + \Psi_Q \cos(QR + \phi)], \quad (2)$$

separates the internal bond form factor f_b from the spatial structure of the condensate. An internal leg/rung sign change alone is not evidence of a pair-density wave. Nor does a boundary-induced modulation of a predominantly uniform pair field establish a distinct finite-momentum superconducting phase [10].

2.2 Two views of stripes and their modern interpretation

In a common-Mott-physics interpretation, singlet and magnetic correlations provide a shared microscopic starting point for several possible ordered states. Static stripes are then one possible competitor, and their presence need not be a prerequisite for pairing [3]. In a stripe-centered interpretation, spatial organization helps create the conditions for pairing. The spin-gap proximity mechanism, for example, couples a conducting stripe to a magnetically correlated environment; pair hopping transmits a spin gap to the mobile carriers, while Josephson coupling acts at a separate stage to establish coherence [27].

These interpretations share an emphasis on strong correlations and singlets. Their sharper disagreement concerns whether stripe organization is an essential intermediate step. The observations of planar superconductivity in stripe-ordered LBCO and the associated layer-decoupling proposal show how static stripes can coexist with pairing while suppressing a particular route to bulk coherence [7, 8]. Conversely, coexistence in a model or material does not establish that stripes caused its pairing.

The relation to the pseudogap remains unresolved. NMR measurements in Eu-LSCO find spin-stripe order extending practically to the pseudogap endpoint and identify competition with superconductivity, providing evidence for a close connection without determining its causal origin [35]. The spin gap of an isolated ladder, an electronic pseudogap in a cuprate, and the superconducting quasiparticle gap are different observables. A calculation of one should not be presented as a measurement or explanation of the others.

2.3 What magnetic spectroscopy establishes

Incommensurate magnetic peaks can be understood through antiphase spin domains. With lattice spacing set to one, a schematic stripe texture is

$$\langle S_{x,y}^z \rangle = m(-1)^{x+y} \cos(\kappa x), \quad q_c = 2\kappa, \quad \mathbf{q}_s = (\pi \pm \kappa, \pi). \quad (3)$$

The charge period is half the period of the staggered spin envelope. This spatial relation is useful evidence for an antiphase stripe texture, but its presence on a finite open ladder does not determine the thermodynamic amplitude or preferred wavelength.

The similarity between magnetic spectra in stripe-ordered LBCO and YBCO was an important motivation for stripe-centered interpretations [6]. Quantum stripe calculations show that low-energy spin-wave-like excitations and higher-energy ladder-like triplet excitations can occur within the same ordered model [36]. Consequently, a failure of a simple semiclassical intensity pattern does not rule out stripes or all low-energy spin waves. The connection between a spectral shape and a microscopic pairing mechanism requires more than qualitative resemblance.

The magnetic ladders in such spin-only constructions represent magnetic regions of a stripe pattern. The present model instead uses doped electronic ladders containing mobile charge. Moreover, its current analyses concern ground-state profiles and equal-time correlations, rather than a computed $S(\mathbf{q}, \omega)$. They can test the static antiphase relation in Eq. (3), but cannot yet demonstrate an hourglass spectrum or identify a neutron resonance.

2.4 Extended Hubbard ladders, materials, and transverse coupling

Magnetic ladder insulators establish the relevance of the ladder building block, but do not by themselves realize a doped electronic array. The spin gap observed in SrCu_2O_3 is an early example [37]. Introducing mobile holes is a separate materials problem: chemical substitution can modify exchange paths, disorder, and carrier localization. Hole doping is possible in $\text{La}_{1-x}\text{Sr}_x\text{CuO}_{2.5}$, where an insulator-to-metal transition was observed without superconductivity down to 5 K in the reported measurements [38]. Its coupled-ladder electronic structure has substantial three-dimensional coupling [39]. Such compounds motivate studying coupled ladders and dimensionality, but do not identify our particular square or cubic kernels with a complete material Hamiltonian.

The superconducting chain-ladder cuprates provide a more direct motivation for the trellis extension. Uehara et al. reported superconducting onset temperatures of 12 K at 3 GPa and 9 K at 4.5 GPa in $\text{Sr}_{0.4}\text{Ca}_{13.6}\text{Cu}_{24}\text{O}_{41.84}$ [23]. The structure contains both Cu_2O_3 two-leg ladder sheets and CuO_2 chains. Neighboring ladders are offset longitudinally, producing the two zigzag paths of a trellis network [40]. Ca substitution is isovalent; it redistributes holes between chains and ladders rather than simply adding a known number of ladder holes. NQR and pressure-dependent optical measurements likewise support pressure-induced transfer of holes into the ladders [41, 42]. Consequently, the model density $n = 15/16$ cannot be assigned directly to a Ca concentration or experimental pressure.

The experimental charge-order landscape is richer than a monotonic conversion from a CDW to a superconductor. Transport, dielectric, and optical measurements found a CDW transition scale decreasing from about 210 K at $x = 0$ to 10 K at $x = 9$ [43]. Resonant x-ray scattering established a ladder hole crystal in $\text{Sr}_{14}\text{Cu}_{24}\text{O}_{41}$ and commensurate charge order also at $x = 10, 11, 12$ [25, 26]. High Ca content alone therefore does not establish the absence of CDW order. Composition, pressure, and commensurability must be specified when comparing a uniform paired model state with experiment.

Static magnetic order and dynamic spin correlations must also be separated. Neutron measurements on $\text{Sr}_{2.5}\text{Ca}_{11.5}\text{Cu}_{24}\text{O}_{41}$ found that the ordered moment decreased under pressure and appeared to disappear near the superconducting threshold [44]. A paired state without a resolved SDW is compatible with that trend, but vanishing static magnetization does not determine the spin gap. High-pressure NMR found gapped quasiparticle excitations in the superconducting state, described as *s*-wave-like in that specific sense [45]. This is not a phase-sensitive measurement of the

relative signs of leg and rung pairing. A ladder's d -wave-like bond structure need not force nodes on its Fermi surfaces; neither agreement nor disagreement of pairing symmetry follows from our bond signs alone. RIXS on Ca-substituted material further motivates including localization: the observed magnetic spectra were interpreted using off-ladder impurity potentials that immobilize holes [46].

Each ladder has L rungs, two legs $\ell = 0, 1$, and $2L$ physical sites. Its Hamiltonian is

$$H_{\text{lad}} = -t \sum_{x=1}^{L-1} \sum_{\ell, \sigma} (c_{x\ell\sigma}^\dagger c_{x+1, \ell\sigma} + \text{H.c.}) - t_0 \sum_{x, \sigma} (c_{x0\sigma}^\dagger c_{x1\sigma} + \text{H.c.}) \\ + U \sum_{x, \ell} n_{x\ell\uparrow} n_{x\ell\downarrow} + V \sum_{\langle a, b \rangle_{\text{lad}}} n_a n_b. \quad (4)$$

The final sum includes nearest-neighbor leg and rung bonds. Thus $V < 0$ is attractive in the density channel. Open boundaries are used along the ladder; $n = N/(2L)$ is the density per physical site and $p_h = 1 - n$ the hole density. The rung hopping t_0 and interladder hopping $t_\perp$ must not be confused with the next-nearest-neighbor hopping t' in two-dimensional Hubbard studies.

An approximate parameter scale comes from the optical analysis of $\text{Sr}_{14}\text{Cu}_{24}\text{O}_{41}$ by Popović et al. [40]: $t = t_0 = 0.26$ eV, interladder hopping $t_{xy} = 0.026$ eV, and $U = 2.1$ eV. Interpreting t_{xy} as the amplitude on each equivalent zigzag path gives $U/t \simeq 8.1$, $t_0/t = 1$, and $\tau_0/t = \tau_1/t = 0.1$, motivating our $(8, 1, 0.1, 0.1)$ comparison. These are effective parameters from an optical model of the parent compound, not a pressure-resolved fit to Uehara's superconducting sample. That analysis does not independently determine the nearest-neighbor V used here; $V = 0$ is our baseline assumption.

There is independent motivation to vary V . The RIXS-DMRG comparison of Padma et al. favors an effective attraction $V/t \simeq -1$ to -1.25 for $\text{Sr}_{14}\text{Cu}_{24}\text{O}_{41}$ [34]. Their model uses $t = 0.38$ eV, $t_0/t = 0.84$, and an additional *intraladder* diagonal hopping $t'/t = -0.3$, distinct from our interladder τ_0, τ_1 . It adjusts U with V to hold the exchange scale fixed, whereas our scans hold $U/t = 8$. Thus its fitted attraction motivates extended interactions without providing a directly transferable parameter set for the present Hamiltonian. Pairing at $V = 0$ in our approximation would not establish that the material has no effective attraction. Chains, explicit oxygen orbitals, structural modulation, disorder, and longer-range Coulomb interactions are also absent from Eq. (4).

The transverse reduction uses the positive scale

$$g = \frac{t_\perp^2}{|E_p|}, \quad E_p^{(h)} = E(N-2, S_z=0) + E(N, S_z=0) - 2E(N-1, S_z=\frac{1}{2}), \quad (5)$$

where negative $E_p^{(h)}$ denotes favorable hole binding. Geometry-dependent kernels convert anomalous, exchange, and density correlators into fields for the next ladder solve. The density is targeted through a chemical potential. The present ansatz retains real, spin-diagonal normal fields and real anomalous fields; it does not provide unrestricted complex phases or noncollinear magnetism (Evidence 1).

For illustration, define $\gamma = 2g$. The same-leg pairing kernels, organized by the two leg indices, are

$$K_\alpha^{\text{square}} = \gamma \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad K_\alpha^{\text{cubic,u}} = \gamma \begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix}, \quad K_\alpha^{\text{cubic,f}} = \gamma \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \quad (6)$$

These matrices summarize one part of the full map, not the entire Hamiltonian. An equation-level comparison identifies the repulsive-ladder benchmark of Ref. [30] with the branch called *cubic*

frustrated in this project. That name is the project's nomenclature. Its published result should not be assigned to all three geometries (Evidence 2).

The trellis adds a longitudinal displacement to the transverse map. Writing m for the ladder index, a minimal hopping Hamiltonian is

$$H_{\text{zigzag}} = - \sum_{m,i,\sigma} c_{m,i,1,\sigma}^\dagger (\tau_0 c_{m+1,i,0,\sigma} + \tau_1 c_{m+1,i+1,0,\sigma}) + \text{H.c.} \quad (7)$$

The symmetric case has $\tau_0 = \tau_1$; removing the shifted path recovers square connectivity in the one-ladder embedding. Triangular connections can frustrate interladder antiferromagnetic alignment, making this geometry a natural test of whether static stripe order remains favored over pairing. The outcome must be calculated with both channels present; it is not implied by frustration alone.

In the same common-denominator approximation, let T_{mn} map sites on neighboring ladder n to ladder m , including both paths, and define $F_{ab} = \langle c_{b\uparrow} c_{a\downarrow} \rangle$ and $X_{ab}^\sigma = \langle c_{a\sigma}^\dagger c_{b\sigma} \rangle$. For the real fields retained here, the kernels take the form

$$\alpha_m = \frac{2}{|E_p|} \sum_{n \sim m} \mathcal{P} [T_{mn} (\mathcal{P} F_n) T_{mn}^T], \quad (8)$$

$$B_{m\sigma} = \frac{2}{|E_p|} \sum_{n \sim m} \mathcal{P} [T_{mn} \mathcal{P} (X_n^\sigma - \frac{1}{2} I) T_{mn}^T]. \quad (9)$$

Here $\mathcal{P}$ retains rung separations within the chosen range cutoff, and B combines diagonal density and off-diagonal normal fields. Each neighbor contributes separately; the two hops of a pair transfer are not split between different neighboring ladders. The shifted paths introduce mixed $\tau_0 \tau_1$ terms, so a coordination-number rescaling of Eq. (6) is insufficient. In particular, diagonal normal fields can depend on bond correlations as well as density (Evidence 15).

Reciprocity fixes the apparent directional asymmetry of the rung indexing. With $A = \tau_0 I + \tau_1 S$ and $S_{ij} = \delta_{j,i+1}$, the one-ladder map contains $A F^{00} A^T$ on leg 1 and $A^T F^{11} A$ on leg 0, with the range projections understood. Opposite shifts belong to opposite spatial neighbors at every iteration; alternating the indexing convention between iterations would instead alternate the map being solved. The comparison therefore uses a fixed reciprocal one-ladder cell and an explicit rectangular two-ladder A/B cell. Both are stationary spatial ansatzes, with different allowed transverse arrangements of longitudinal textures and different open end cuts. The two-ladder calculation targets the same mean density on each ladder separately; it does not allow unequal average fillings (Evidence 15).

These runs test whether geometry can change the selected order at fixed intraladder parameters and $V = 0$. A paired solution without resolved bulk CDW/SDW would be qualitatively compatible with a superconducting regime of the chain-ladder family, but would not locate it in pressure or reproduce the material's whole phase diagram. The first available trellis trajectory and its unresolved convergence and cell-comparison checks are described in Section 3.11. Its anomalous amplitude is not a measurement of T_c , phase stiffness, or a dynamical spin gap.

This construction is conceptually related to coupled-ladder pairing mechanisms but does not implement every assumption of the original stripe proposals. In particular, the explicit alternating dopings in the period-four construction of Ref. [28] differ from an array of equivalent ladders, although that work also considers an equivalent-ladder period-two limit.

We study equivalent coupled ladders as a prescribed microscopic geometry. The observed stripe texture is an additional longitudinal charge and spin modulation; within the repeated-ladder embedding, its domain walls extend transverse to the ladder axes. This transverse alignment is a

restriction of the embedding rather than an independently optimized arrangement of distinct ladders. Collective transverse quantum fluctuations are approximated by static mean fields.

3 Results and interpretation accumulated to date

3.1 Evidence hierarchy and common parameters

The strongest current comparison concerns the square geometry at $t_0/t = 1.4$, $t_\perp/t = 0.1$, $U/t = 8$, and $n = 15/16$, with $V/t = 0$ and -0.4 . The latest paired and striped starts use $L = 64$ rungs and bond dimension $\chi = 200$. A separate $V = 0$ comparison tests $\chi = 400$, and the isolated-ladder backbone reaches $\chi = 1200$. These are different calculations with different error controls, rather than one uniformly converged data set.

It is useful to distinguish an observed texture, a numerically self-consistent state, stability against specified perturbations, and a thermodynamic ground state. The recent trajectories provide strong texture and basin information. Their remaining acceptance failures and the incomplete length and bond-dimension studies prevent the last two categories from being identified without qualification. Here, *basin* describes access under a specified numerical map, not a measured free-energy basin.

3.2 The isolated ladder already has substantial pairing tendencies

The six-sector backbone at $V = 0$ and $t_0/t = 1.4$ gives the finite-system quantities in Table 1. All six final $\chi = 1200$ sectors pass their stored numerical gates. Changes from $\chi = 800$ are useful sensitivity observations, but not all sectors pass at that earlier stage; the comparison therefore does not constitute a complete bond-dimension extrapolation (Evidence 3, Evidence 5).

Table 1: Isolated-ladder evidence at $L = 64$, $U/t = 8$, $n = 15/16$, $t_0/t = 1.4$, and $V = 0$, from the final $\chi = 1200$ backbone. Negative binding energies use Eq. (5)’s convention.

Quantity	Value	Interpretation
Spin gap Δ_s/t	0.1533613	Fixed-particle-number spin excitation
Even-particle charge gap $\Delta_{c,2}/t$	0.01085830	Finite-size charge curvature
Hole binding $E_p^{(h)}/t$	-0.1465102	Favorable two-hole binding
Particle binding $E_p^{(e)}/t$	-0.1485427	Favorable two-particle binding
Pair decay exponent	0.7614 ± 0.0215	Fit to rung-pair correlations
Charge decay exponent	1.2158 ± 0.0830	Less reliable fit to charge correlations

The measured pair correlations decay more slowly than the fitted charge correlations. Pair fits have R^2 between 0.953 and 0.982, while charge fits give 0.527–0.707. The quoted fit errors do not include the full finite-size, boundary, and truncation uncertainty. These results are consistent with a spin-gapped ladder with appreciable pairing correlations, but do not by themselves establish its asymptotic exponents or the ordered state of an array [21, 22].

Two other diagnostics illustrate the importance of conventions and fit quality. The stored charge parameter is 1.4247 with rung normalization and 0.7124 with site normalization. An unqualified symbol K_ρ would therefore be ambiguous. A central-charge fit yields approximately 2.009 with $R^2 = 0.155$ and is excluded from physical interpretation. The open-boundary density profile has a

strong component at $q/\pi = 1/8$, which can arise as a Friedel response and is not independent proof of spontaneous charge order [47].

The Stage 1 equal-time covariance spectra are broad, with participation ranks around 48–63. The first bulk spin-odd candidate follows two edge-localized modes, and a charge mode at $q/\pi = 1/8$ first appears at rank 11. This motivated combining physically chosen fields with covariance-based directions instead of assuming that a handful of leading covariance modes would exhaust the relevant response. Equal-time covariance is itself distinct from a static susceptibility, which also depends on excitation energies.

3.3 Response discovery: a strong pairing direction with unresolved controls

The Stage 2 pilot reduces fourteen named fields to twelve independent probe directions: nine normal and three pairing directions. Its projected mean-field response has a leading square pairing multiplier of about 4.6688 and a leading positive normal multiplier of 1.1879. The cubic cases both give a leading pairing multiplier of 24.4730, while their leading normal behavior differs: the largest-magnitude normal multiplier is approximately -1.1879 for the frustrated construction and $+3.5637$ for the unfrustrated one (Evidence 4).

These values show that the isolated ladder supplies strong candidate pairing response and that transverse geometry changes the competing normal channels. They do not establish which nonlinear state wins. The discovery reciprocity error is approximately 0.121% and the normal/pair cross-block ratio 0.332%, but the maximum omitted-response norm fraction is 0.83267. Small errors within the projected matrix can therefore coexist with a large response outside the chosen subspace. The second-amplitude validation was not present in the reviewed evidence (Evidence 5).

More fundamentally, the zero-field isolated ladder is not a fixed point of the full normal mean-field map: its square-map image has a normalized field norm of approximately 0.0306655, dominated by normal fields. A derivative at that reference is a bare-ladder diagnostic, not the stability matrix of a self-consistent dressed normal state. The associated square estimate $t_{\perp}/t \simeq 0.04628$ must consequently remain a projected bare-reference onset estimate, not a phase boundary.

This resolves the apparent tension between a large bare pairing response and the later nonlinear stripe trajectories. They concern different reference states, finite-amplitude feedback, and restricted response information. Positive or negative multipliers describe growth or alternation of the specified iteration. Neither is a physical excitation frequency or a global energy criterion. A physical two-sublattice interpretation of a raw two-cycle also requires a mapping to the transverse lattice and the appropriate partner-field energy [31].

3.4 Why the earlier paired-state interpretation was revised

The loose six-seed $V = 0$ campaign initially produced six software-accepted, pairing-bearing endpoints, with a corrected energy spread of only $6.0274 \times 10^{-7}t$ per physical site. This suggested a common paired region, but the later history analysis showed that it did not establish a unique stable paired basin. Three histories stopped after six evaluations; three others showed coherent spin-density growth during raw iteration and subsequent suppression under Anderson acceleration. The conspicuous change in one trajectory is reproduced by its first Anderson coefficients, $+22.6811$ and -21.6811 (Evidence 7).

This finding does not imply that Anderson acceleration changes the fixed-point equation or that every accelerated solution is unphysical. It shows that short histories and accelerated access

do not test stability against a resolved competing-order perturbation. The subsequent 95%/5% two-family experiment addresses that question by using representative paired and striped reference fields, adding a finite component of the competing order, and retaining raw updates throughout.

A related warning comes from the square $(t_0/t, V/t) = (1.0, -0.4)$ coverage point. Its *diverging* outcome is a convergence classification: the fields remain bounded while pairing gives way to a stripe-like texture and an accelerated residual spike triggers the stop. It is not evidence of a divergent physical observable or an accepted phase (Evidence 8).

3.5 Attraction: two contrasting starts approach a common paired state

At $(t_0/t, V/t) = (1.4, -0.4)$, both original 80-evaluation trajectories approach the same pairing-dominated state. The initially strong stripe’s bulk leg-odd spin Hartree RMS decreases from about $1.99 \times 10^{-2}t$ to $3.88 \times 10^{-8}t$; the pairing start ends near $2.49 \times 10^{-8}t$. Both pairing-field RMS values settle near $2.8236 \times 10^{-3}t$. Bulk RMS here uses rungs 6–59 (Evidence 9).

The physical symmetrized anomalous correlators have bulk leg amplitudes near $+0.035217$ and rung amplitudes near -0.053242 , establishing a *d*-wave-like internal sign pattern in the stored convention. These are not normalized versions of Eq. (1) or pair-pair correlators. The full anomalous matrices agree to approximately 5.85×10^{-5} in relative L2 norm, and the largest site-density difference is below 6.23×10^{-6} . The residual boundary density profile remains visible; its survival does not indicate a surviving strong static spin stripe.

The terminal corrected energies differ by only $7.49 \times 10^{-9}t$ per site, supporting agreement of the two trajectories at their resolved accuracy. Both historical acceptance flags remain false. The principal issues are residual fluctuations in nearly vanishing channels, an oversensitive scalar slow-mode check, and a marginal energy-window miss in one lineage. The later noise-aware history checks support the plateau interpretation, but do not retroactively certify every original gate. Thus the result is strong evidence for a common paired attractor at these numerical settings, without proof of a global minimum or higher- χ stability.

3.6 Zero attraction: stripe order grows while anomalous pairing recedes

The corresponding $V = 0$ trajectories behave differently. The stripe start reaches 80 evaluations with bulk leg-pairing field RMS $1.75 \times 10^{-10}t$ and spin Hartree RMS $0.02414t$. The pairing start ends at the wall-time limit after 62 evaluations: its pairing RMS has fallen from $0.004502t$ to $0.0007090t$, while spin RMS has grown from $0.001816t$ to $0.02251t$. Pairing falls by another 65.6% across the final five records. The trend is already present before the deadline-limited final density solve (Evidence 10).

The residual anomalous correlators in the pairing-start trajectory still have opposite leg/rung signs, but their amplitudes are decreasing rapidly. They do not establish settled coexistence. The two spin textures are similar but not identical: the bulk physical spin profiles differ by 26.4% in relative L2 norm, and the density difference can reach 0.04365 per site. The terminal global relative residuals are about 5.88×10^{-4} and 2.65×10^{-2} for the stripe and pairing starts, respectively, against a 10^{-4} threshold.

The working interpretation is that a robust stripe texture has been identified and that the finite competing-order perturbation drives the pairing-dominated start toward it under raw iteration. Strict spatial self-consistency remains pending. In particular, the stripe amplitude’s apparent saturation does not eliminate slow rearrangement of the profile. These observations support strong

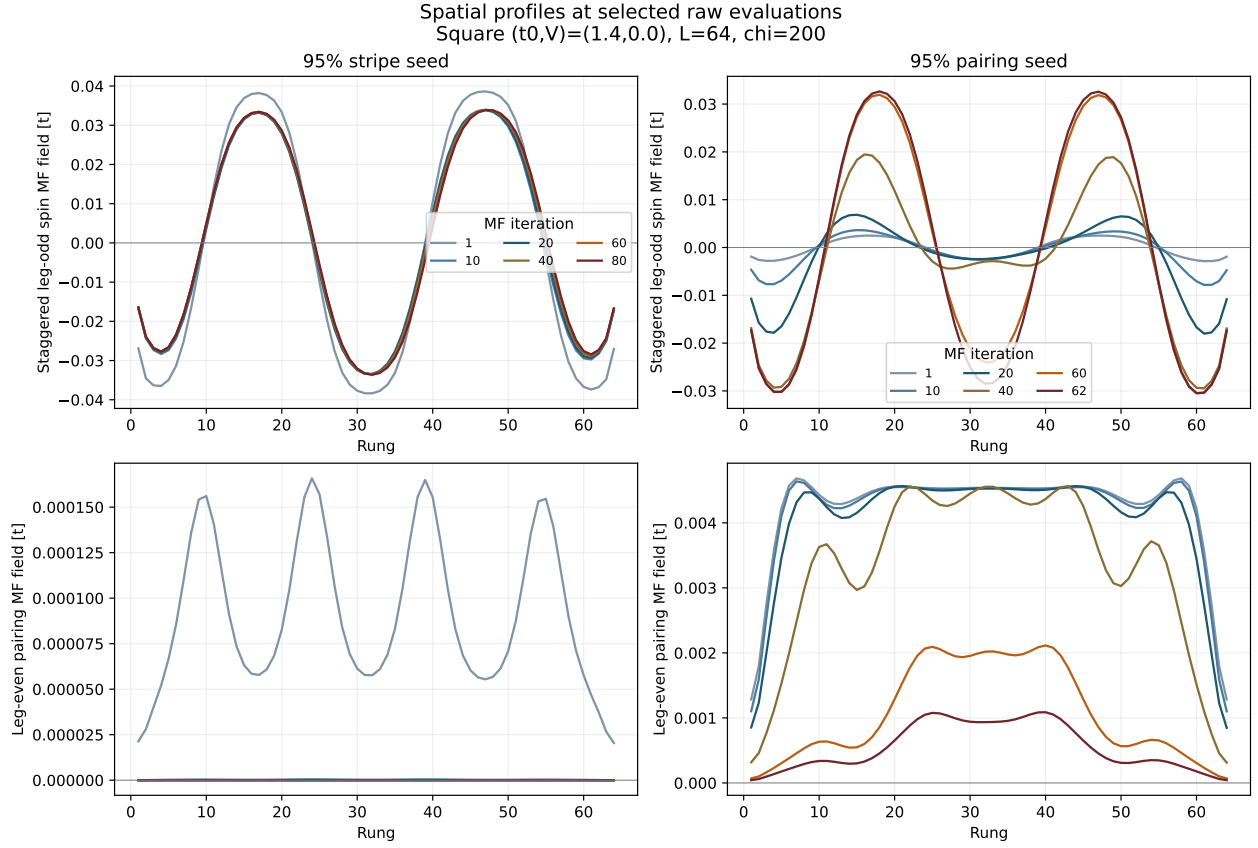


Figure 1: Previously generated spatial-history figure for the two $V = 0$ anchor trajectories. The upper row shows the staggered leg-odd spin mean field; the lower row shows the leg-even pairing mean field. The two lower panels use different vertical scales. The stripe start loses pairing, while the pairing start develops a stripe texture and its pairing decreases. Curves are self-consistency evaluations, not physical time evolution. The final states remain unaccepted for resolved field drift. Source: [Evidence 10](#).

competition between static spin/charge order and the anomalous pairing solution at this point. They do not prove that the final stripe is the global minimum or that all local pairing correlations have disappeared.

3.7 Stripe wavevectors and slow texture relaxation

Both $V = 0$ endpoint profiles have dominant full-length Fourier modes $m_c = 4$ for charge and $m_s = 30$ for leg-odd spin. With $q_m = 2\pi m/L$ and $L = 64$,

$$q_c = \frac{\pi}{8}, \quad q_s = \frac{15\pi}{16}, \quad \kappa = \pi - q_s = \frac{\pi}{16}. \quad (10)$$

The observed relation $q_c = 2\kappa$ and the sign reversals of the staggered spin envelope near charge-depleted regions support an antiphase stripe interpretation. The charge period is 16 rungs and the spin-envelope period 32 rungs.

At hole density $p_h = 1/16$, the magnetic displacement in reciprocal-lattice units is $\delta_s = \kappa/(2\pi) = 1/32 = p_h/2$. It differs from the often discussed cuprate relation $\delta_s \simeq p_h$. In an ideal domain-wall counting picture with stripe filling ν holes per unit transverse length, $p_h = \nu/d$ and $\delta_s = 1/(2d)$; the present wavevectors would correspond to $\nu = 1$, rather than the $\nu = 1/2$ used in the usual half-filled-stripe cartoon. This is a conditional interpretation of the finite-system wavelength, not a measurement of wall filling. Neither the preferred wavelength nor its scaling with doping has been established.

The September 13 follow-up explains why the stripe looks settled while failing strict convergence. Its bulk spin RMS changes by only 0.0528% over the last twenty records, but more than 99.999% of the squared final spin residual is orthogonal to an overall rescaling of the applied spin field. Approximately 99% of that residual power lies in the bulk. The dominant remaining error concerns spatial shape.

For example, interpolated crossings of the staggered spin envelope near the third and fourth walls shift from 39.3999 and 55.1830 at evaluation 60 to 39.5165 and 55.2673 at evaluation 80. Unequal shifts reveal texture rearrangement. They are sub-rung profile diagnostics, not measured motion of physical objects. The stopping diagnostic estimates a contraction near 0.99059, suggesting very slow relaxation if that behavior persists; the extrapolation is neither a certified error bound nor a physical soft-mode eigenvalue (Evidence 10).

3.8 Energy comparisons and what the bond-dimension check adds

The relevant energy is the canonical variational functional, including the mean-field double-counting contribution. If L_{lin} denotes the expectation of the applied linear mean-field terms,

$$E_{\text{var}} = E_{\text{eff}} + \mu N + (E_{\perp} - L_{\text{lin}}) = \langle H_{\text{lad}} \rangle + E_{\perp}. \quad (11)$$

The density-targeted diagnostic is $E_{\star} = E_{\text{var}} + \mu(N_{\star} - N)$, a leading correction to the target density. It does not remove curvature and optimization errors. Effective-Hamiltonian eigenvalues alone cannot rank the branches (Evidence 1).

At $V = 0$, the separate matched $\chi = 400$ comparison gives a stripe endpoint lower than the paired endpoint by $2.4336 \times 10^{-4}t$ per physical site, or $0.03115t$ on the 128-site ladder. The September 6 working decision treated the available evidence as sufficient to regard the advantage as reasonably robust over the tested $\chi = 200$ to 400 increase. Neither original endpoint satisfies all

formal acceptance gates, so the value remains a terminal diagnostic difference rather than a fully certified fixed-point ranking (Evidence 6).

The newer raw $\chi = 200$ trajectories instead end with a separation $8.99 \times 10^{-5}t$ per site while the pairing lineage is still evolving. This is not an independent estimate of the same equilibrium energy gap. Similarly, an early low energy on an unconverged trajectory cannot be selected as a lower-energy phase. During startup, the functional combines the applied partner fields with current correlations; raw self-consistency iteration is not guaranteed to minimize a single self-consistent state’s energy monotonically.

3.9 Complete square and unfrustrated-cubic grids

The September 18 review extends the anchor comparison to $t_0/t = 1.0, 1.2, 1.4$ and $V/t = 0, -0.2, -0.4$, with $U/t = 8$, $n = 0.9375$, $t_\perp/t = 0.1$, $L = 64$ rungs and $\chi = 200$. The two starts are reciprocal 95%/5% mixtures of the reference stripe and paired correlations, reconstructed with the target couplings. They evolve with raw MF updates, without Anderson acceleration. All phase descriptions below refer to finite-run physical correlations, not certified minima (Evidence 11).

Figure 2 places the two phase diagrams side by side with the same coordinate axes. After including both $(1.4, 0)$ continuations, the coarse square grid has seven stripe coordinates and two paired coordinates, $(1.4, -0.4)$ and $(1.4, -0.2)$. Both seed families now reach the same qualitative basin at each coordinate. At $(1.4, 0)$ the stripe and pairing lineages contain 100 and 82 cumulative evaluations, respectively; the late pairing collapse seen in the earlier history is therefore followed by a stripe endpoint. The whole square grid contains 902 evaluations across 18 lineages, including 20 source jobs, with no formally accepted endpoint (Evidence 12).

In contrast, all 18 unfrustrated-cubic starts reach essentially unpaired CDW/SDW stripes after 60 evaluations. Physical leg-odd spin RMS ranges from 0.280 to 0.356, while the largest leg-pair RMS is 2.21×10^{-11} . The dominant charge and spin modes are $m = 4$ and 30, corresponding to nominal 16-rung charge and 32-rung staggered-spin-envelope periods. The two square paired coordinates are striped in cubic. Every cubic endpoint passes its final energy-window, density, inner-DMRG and identity checks; remaining field/profile and slow-mode checks prevent acceptance. This is evidence of geometry-dependent nonlinear basin selection, not an energetic ranking of different Hamiltonians or an exclusion of every other cubic basin.

The full energy histories in Fig. 3 retain the startup and delayed conversions. In particular, square $(1.2, -0.4)$ has an extended pairing-seed energy plateau before its late decrease; square $(1.4, 0)$ evolves slowly into its continuation. Cubic energy histories generally settle earlier. All plotted energies use the stored canonical functional with target-density correction, and each panel keeps its own vertical scale. The comparison is between numerical trajectories under different geometries, not a ranking of their absolute energies as phases of one Hamiltonian. Early off-self-consistent energy dips and late flat energies do not establish stationary minima.

Figures 4 and 5 show the complementary physical spin and pairing RMS histories. They use the same definitions and logarithmic scales in both geometries, unlike the older standalone square plots of MF-field amplitudes. Spin is the RMS of the physical leg-odd spin on rungs 6–59; pairing is the physical leg-even nearest-neighbor leg-pair RMS on bonds 6–58. Square spin decays while pairing survives at $(1.4, -0.4)$ and $(1.4, -0.2)$; cubic instead retains strong spin and loses pairing at both coordinates. At square $(1.2, -0.4)$ and $(1.4, 0)$, delayed pairing collapse accompanies increasing spin. The long transient in these panels explains why a short, flat energy history can be misleading. Tiny pairing fluctuations after collapse are numerical remnants, not resolved renewed order.

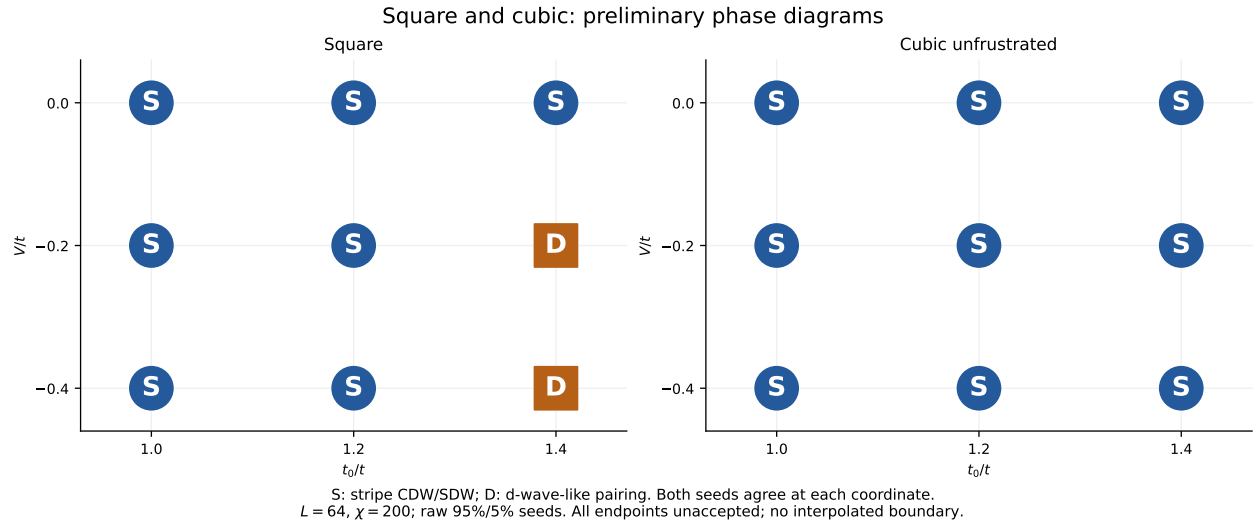


Figure 2: Preliminary square (left) and unfrustrated-cubic (right) phase diagrams. Both seeds reach the same observed family at every coordinate: seven stripe and two paired points on square, nine stripe points in cubic. S denotes CDW/SDW stripes and D opposite-sign leg/rung pairing. All endpoints remain unaccepted at $\chi = 200$; no boundary is interpolated. Sources: [Evidence 12](#) and [Evidence 11](#).

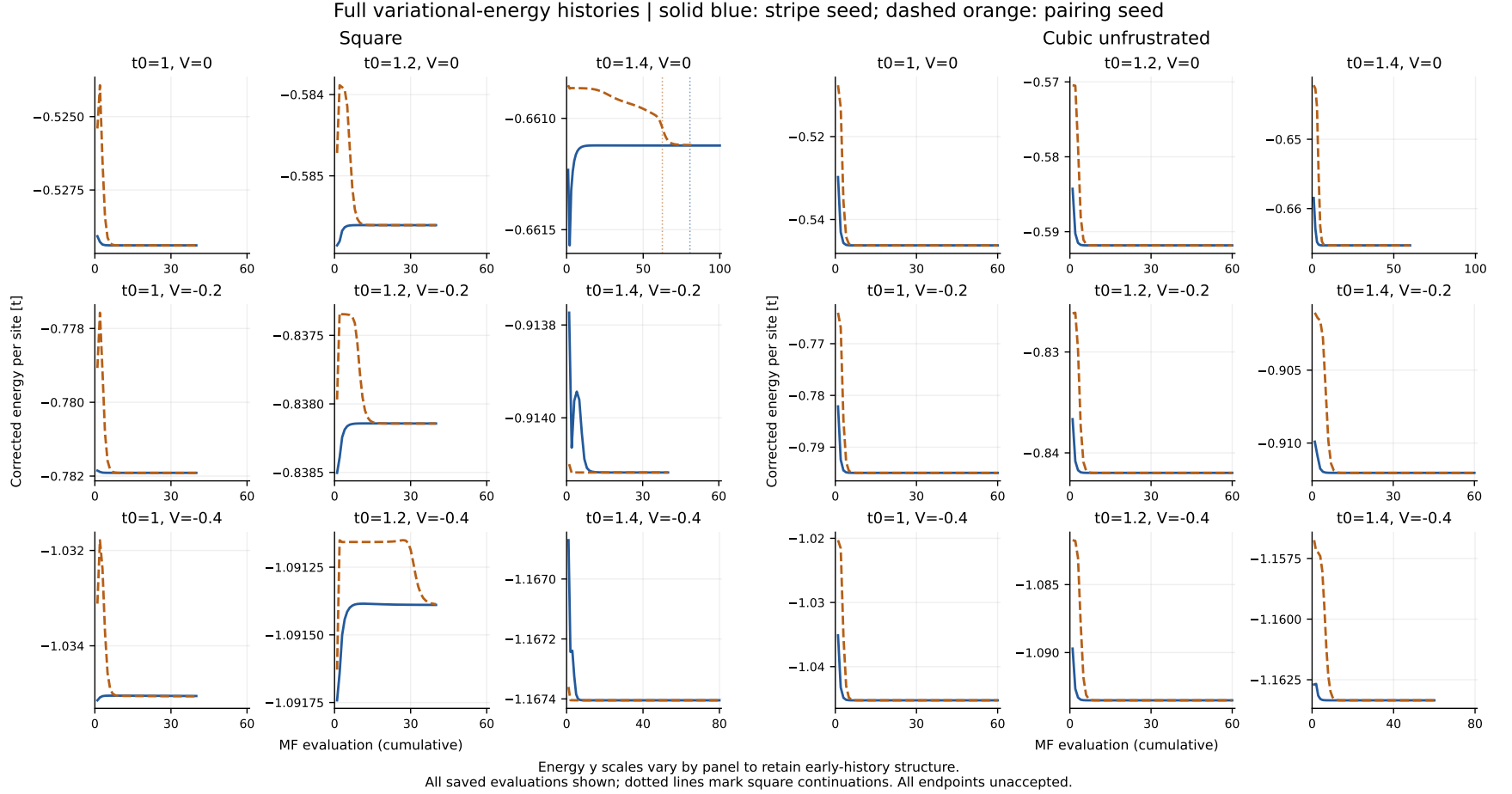


Figure 3: Full target-corrected canonical energy grids, square left and cubic unfrustrated right. Within each geometry, columns are $t_0 = 1, 1.2, 1.4$ and rows are $V = 0, -0.2, -0.4$. Solid blue and dashed orange denote stripe and pairing seeds. All 902 square and 1080 cubic saved evaluations are included. Dotted lines mark the square continuations after evaluations 80 and 62; curves end at their own final evaluations. Matching coordinates share an iteration range, while energy y scales vary by panel.

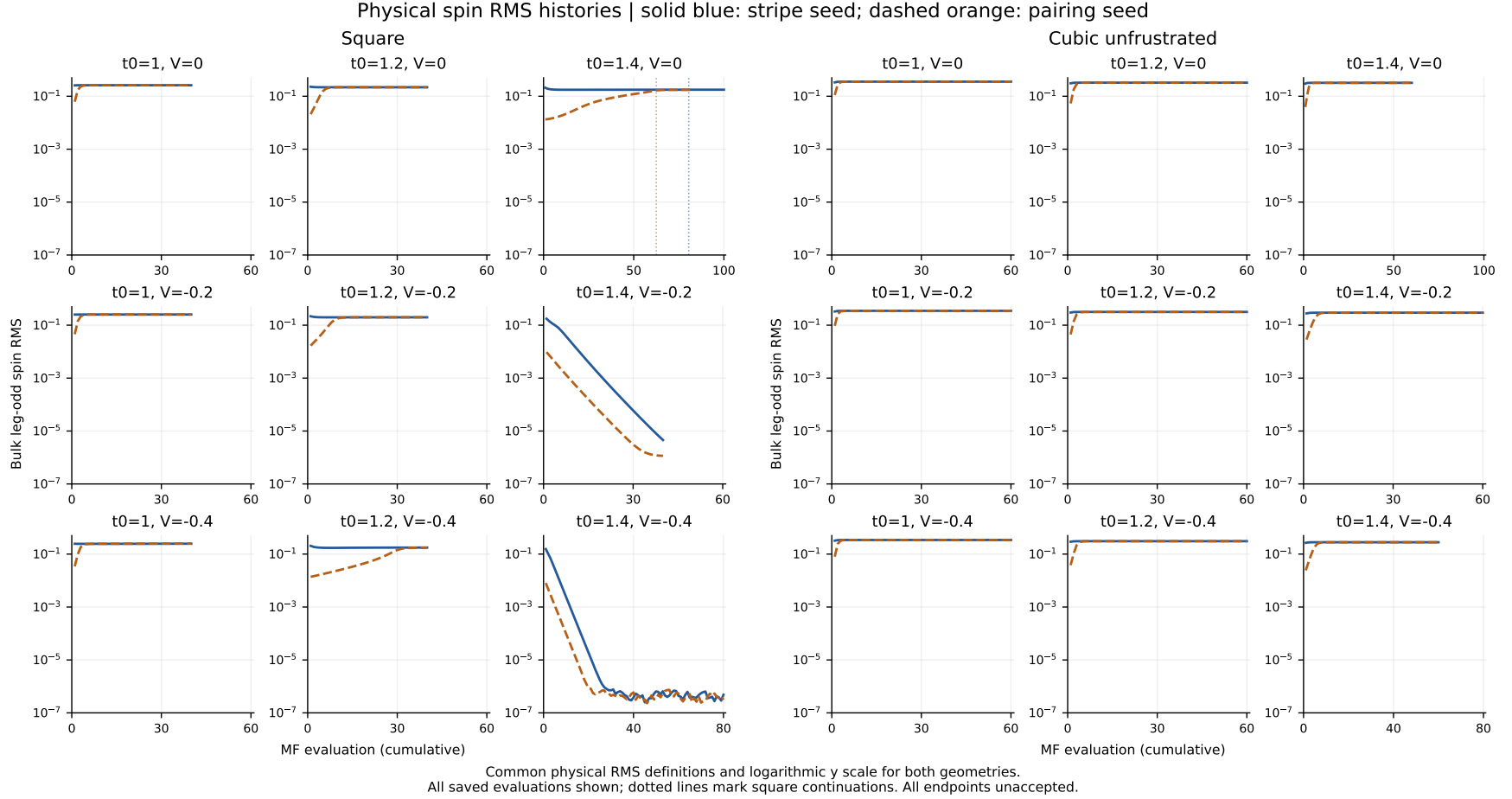


Figure 4: Physical leg-odd spin RMS histories, square left and cubic right, over runs 6–59. All panels use a common logarithmic y scale. The point order, seed colors and continuation markers match Fig. 3. These are physical correlations, not geometry-dependent Hartree-field amplitudes. Spin decays at the two square paired coordinates but remains strong throughout cubic.

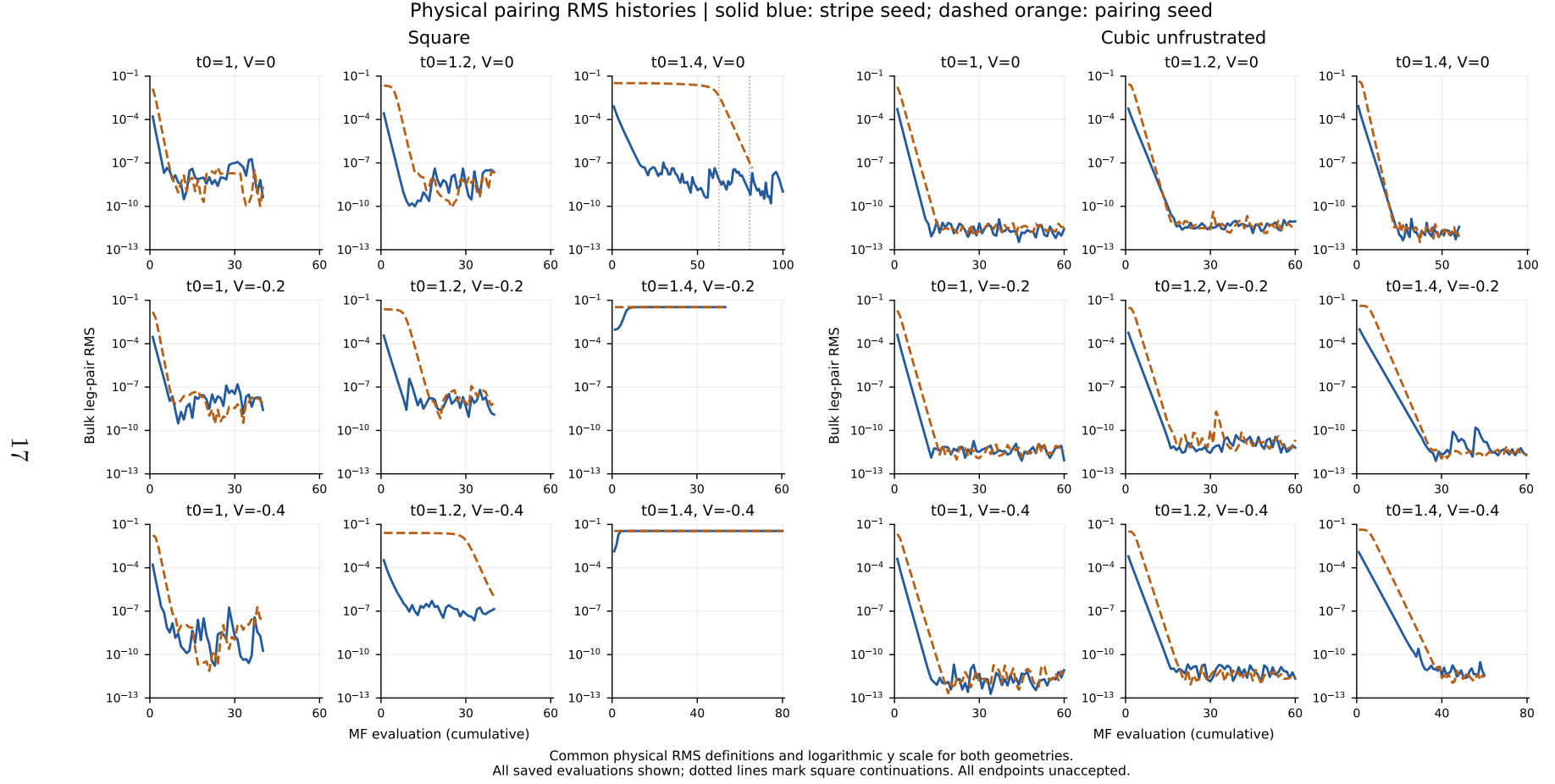


Figure 5: Physical leg-pair RMS histories on bonds 6–58, square left and cubic right, with common logarithmic y scale. The layout matches Fig. 4. Pairing survives at the two square paired points and decays to negligible amplitudes at every cubic point. The full histories retain delayed square pairing collapse, including the $V = 0$ continuation. None of the endpoints is formally accepted.

3.10 Finer square cuts and the shape of the variational energy

The finer cuts add $V/t = -0.15, -0.10, -0.05$ at $t_0/t = 1.4$ and $t_0/t = 1.25, 1.30, 1.35$ at $V/t = -0.4$, with 60 raw evaluations per seed. Both seeds reach paired trajectories at four new points. Distinct stripe and paired trajectories survive at $(1.25, -0.4)$ and $(1.4, -0.05)$. Their signed endpoint energy differences, pairing seed minus stripe seed, are -3.189×10^{-5} and $-5.952 \times 10^{-5} t/\text{site}$. These exceed the sums of the corresponding final-ten energy ranges, 1.574×10^{-7} and $7.461 \times 10^{-8} t/\text{site}$, but recent drift does not bound the remaining convergence error. All twelve endpoints remain unaccepted (Evidence 13).

Figure 6 shows the full stored variational energy along both cuts, including the coarse endpoints and the latest $V = 0$ continuations. The quantity is the canonical functional with its field-dependent double-counting terms and target-density correction,

$$e_{\text{target}} = E_{\text{var}}/N_s + \mu(n_{\text{target}} - n), \quad N_s = 2L = 128. \quad (12)$$

It is not the effective-Hamiltonian eigenvalue. No additional field-independent perturbative offset has been introduced. On this scale the two seed curves nearly overlap: the V cut is almost linear, while the t_0 cut has gradual curvature. Curves identify the initial seed, not a persistent phase, because independent runs can change their outcome as the parameter changes.

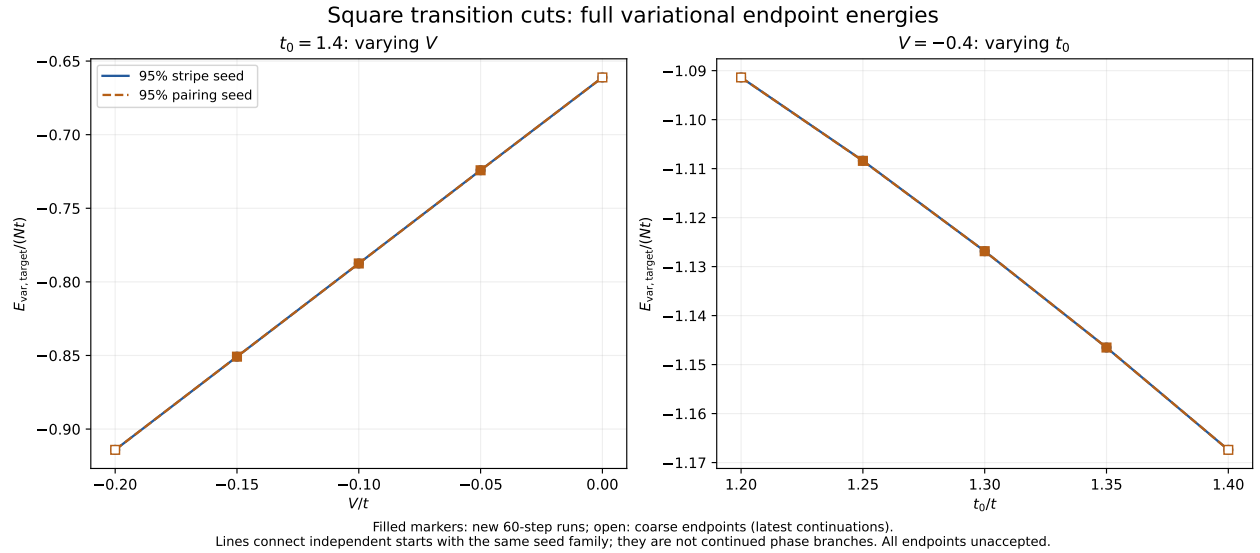


Figure 6: Full canonical variational endpoint energies, corrected to the target density. Filled points are the new 60-step starts; open points use the coarse endpoints and latest continuations. Lines guide the eye between independent starts; they are not parameter-continuation branches. Total energy equals the displayed energy per site times 128.

To resolve small shape changes, Fig. 7 subtracts one common linear chord from both seed curves on each cut. Its endpoints are the mean seed energies at the two outer coordinates. This preserves both seed gaps and changes in slope. The lower panels display adjacent secants $\Delta e/\Delta p$, with $p = V$ or t_0 , rather than a fitted derivative. Along the pairing-seed V curve the slopes are approximately 1.266231, 1.266144, 1.266212 and 1.261367. Thus the last interval, $-0.05 \leq V/t \leq 0$, has a slope smaller by about 0.38% than its predecessor. The stripe-seed curve also bends downward in that interval. This small feature lies near the change from paired to stripe outcomes and is compatible with a weak first-order crossing, but does not resolve one.

For a first-order transition the minimizing energy can remain continuous while its first derivative jumps. Five sampled coordinates do not distinguish that kink from a continuous change of curvature. On the t_0 cut the pairing-seed secants, -0.340515 , -0.368857 , -0.393241 , -0.417749 , change nearly regularly, with no obvious sharp cusp at this resolution. The broad arch after linear subtraction is not itself a singularity. Absence of a visible kink does not exclude a weak first-order transition.

The two distinct textures at one point on each cut make metastability and first order worth testing. However, these are not two stationary phase branches traced through a crossing, nor parameter-sweep hysteresis. The signed E_p values at the new points were linearly interpolated between the coarse endpoints as requested; $g = t_\perp^2/|E_p|$ therefore also changes along each cut. The plotted slopes include this coupling dependence and cannot be identified with the bare V or rung-hopping conjugate observable alone. Coarse and fine runs also have different stopping controls.

The weak spin at the paired $(1.4, -0.05)$ endpoint requires a spatial qualification. Its usual spin RMS on rungs 6–59 rises 1.50% over the last ten evaluations to 0.005606, but 96.7% of its spin-squared weight lies in the outer 14 rungs at each end. Central 32-rung spin still decreases, while leg-pair RMS remains 0.032991. This is not established bulk coexistence or a demonstrated bulk return to stripes. At paired $(1.25, -0.4)$, the spin remainder decreases by 52.1% over the final ten evaluations.

Figure 8 places spin and pairing alongside the energy cuts. Coarse and fine points use the same physical definitions: $M_i = (\langle S_{i,0}^z \rangle - \langle S_{i,1}^z \rangle)/2$, with spin RMS over rungs 6–59, and symmetric nearest-leg anomalous pairing averaged over the two legs before its RMS on bonds with left rungs 6–58. These are not source-field norms or four-fermion pair-pair correlations. Central-window spin exposes the end-weighted remainder at $(1.4, -0.05)$. Pairing stays finite almost up to the sampled change of outcome, while the stripe-seeded amplitudes change at a different sample. This motivates testing competing finite-amplitude branches, but joining independent starts does not measure a discontinuity, coexistence interval or hysteresis. A logarithmic companion in the report resolves the small remainders.

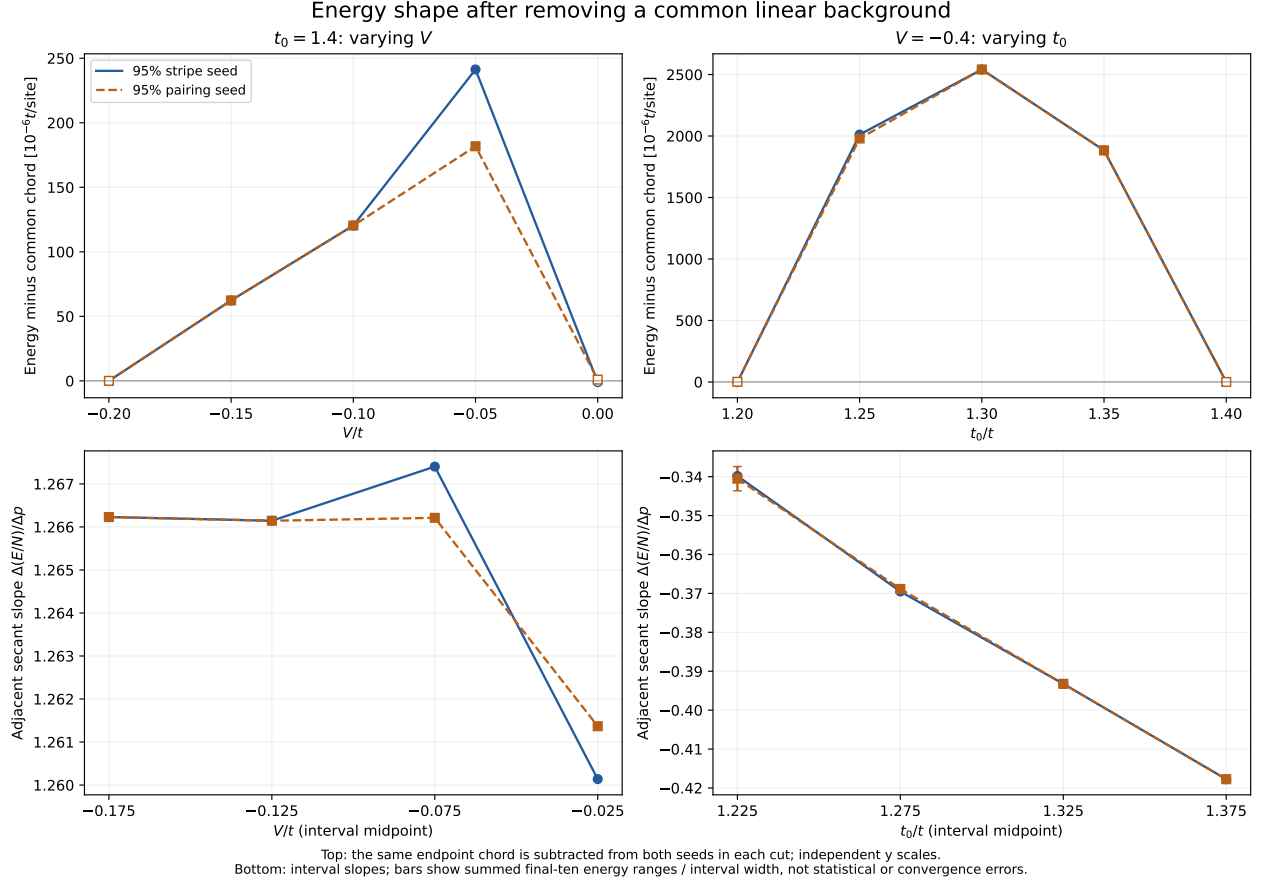


Figure 7: Energy shape and adjacent-interval slopes. The same chord is subtracted from both seed curves within each cut; the two cuts have independent vertical scales. Bars on the secants are the sum of endpoint final-ten energy ranges divided by interval width, not statistical or convergence error bounds. In particular, the coarse $t_0 = 1.2$ pairing lineage still has an energy range $1.56 \times 10^{-4} t/\text{site}$. Polygonal corners between samples are not measured derivative jumps.

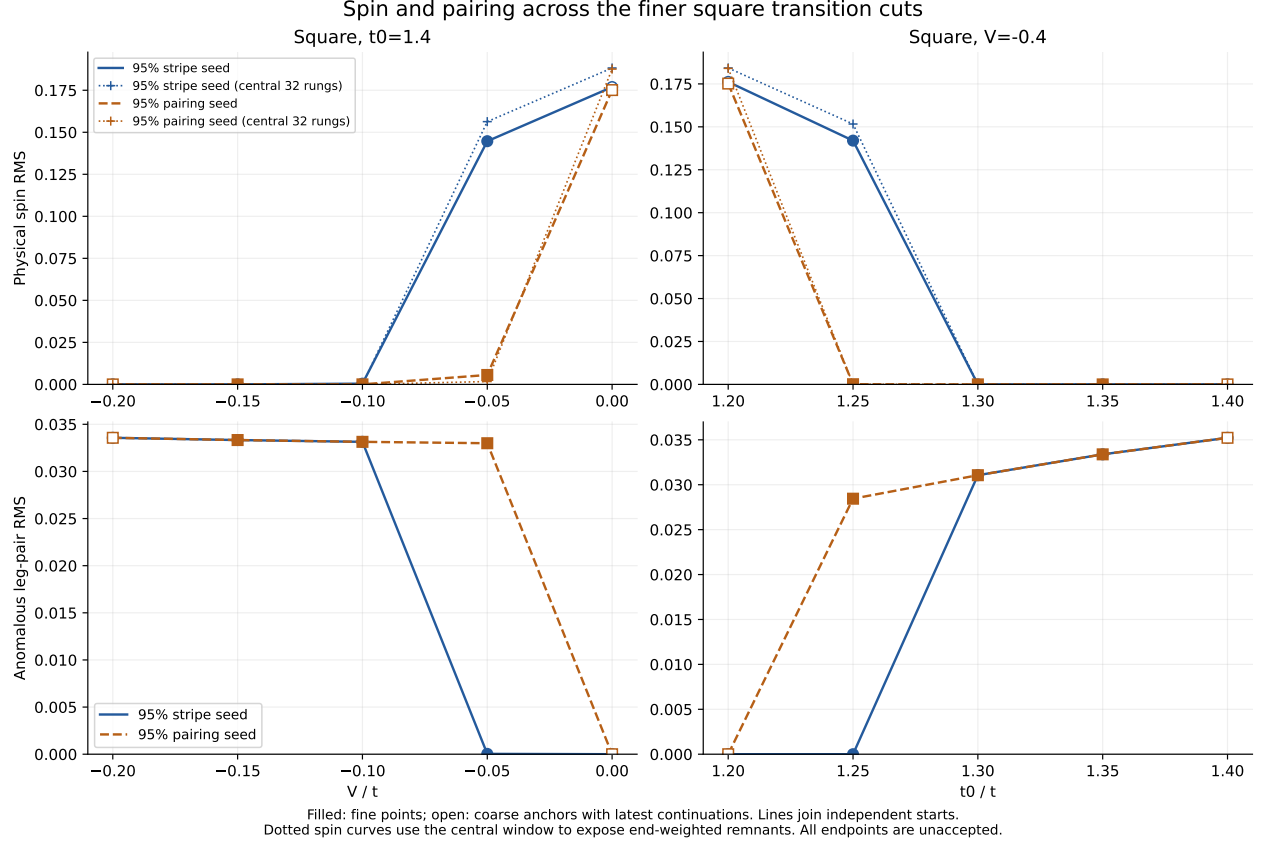


Figure 8: Spin and anomalous leg-pair RMS along both finer square cuts, using the same twenty endpoints as Fig. 6. Filled markers denote fine starts; open markers denote coarse anchors with latest continuations. Dotted spin curves use central rungs 17–48. Lines join independent finite-iteration starts, not stationary phase branches. All endpoints remain unaccepted.

3.11 Repulsive interaction on square: loss of pairing and magnetic defects

All four square $(t_0/t, V/t) = (1.2, +0.2)$ starts now complete 60 raw evaluations at $L = 64$, $\chi = 200$: the two 95%/5% reference mixtures and intertwined patterns with initial charge wavelengths eight and sixteen. The additional seeds place pairing near hole-rich regions and spin nodes, testing whether the legacy frustrated-cubic texture survives on square. Figure 9 shows that all four lose anomalous pairing. Stripe, pairing and period-16 starts end at spin RMS 0.23321–0.23358 and leg-pair RMS below 8.3×10^{-10} . The period-eight start ends at spin RMS 0.20700 and leg-pair RMS 7.01×10^{-9} , with the latter falling another 87.8% over the last ten evaluations. None retains appreciable anomalous coexistence (Evidence 14).

The shorter-wavelength start has not merged with the other three. Figure 10 shows four spin-envelope nodes in the stripe, pairing and period-16 endpoints, near rungs 10, 25, 40 and 55. Their charge/spin Fourier bins are $m = 4/30$, giving nominal charge and spin-envelope periods 16/32. The period-eight start evolves from eight nodes at evaluation one to six near 10.16, 23.21, 26.81, 38.30, 41.89, 54.91. Two close node pairs survive around the inner hole-rich regions. Charge still has dominant $m = 4$, while the spin spectrum has comparable $m = 31, 30, 28$ amplitudes 0.09264, 0.08450, 0.06761. Its largest bin should not be interpreted as a clean new spin wavelength. The spatial evidence is consistent with incomplete coarsening or a trapped magnetic defect pattern, rather than a surviving period-eight intertwined state.

The period-eight energy is $-0.332811625875 t/\text{site}$, versus approximately $-0.333744 t/\text{site}$ for the other three. Its excess over the stripe-seeded endpoint is $9.3242 \times 10^{-4} t/\text{site}$; their summed final-ten energy ranges are $8.96 \times 10^{-8} t/\text{site}$. This is a useful diagnostic distinction, not a certified metastable branch energy. All four fail global field/slow-mode and spatial charge, spin and exchange checks. Maximum pointwise spin changes over the last ten evaluations remain 0.00358–0.00655, despite nearly constant RMS. Stripe and period-eight starts also fail an inner-DMRG window; period sixteen fails the energy window. Flat energy and a steady norm conceal moving walls.

The completed wavelength test strengthens the qualification on the earlier physical interpretation: repulsive V alone does not guarantee intertwining across these geometries. Other square parameters or basins remain possible. Loss of anomalous order also does not determine whether connected pair-pair correlations survive in the stripes. The prepared retrospective measurement pass addresses that separate question; its results are not part of this endpoint analysis.

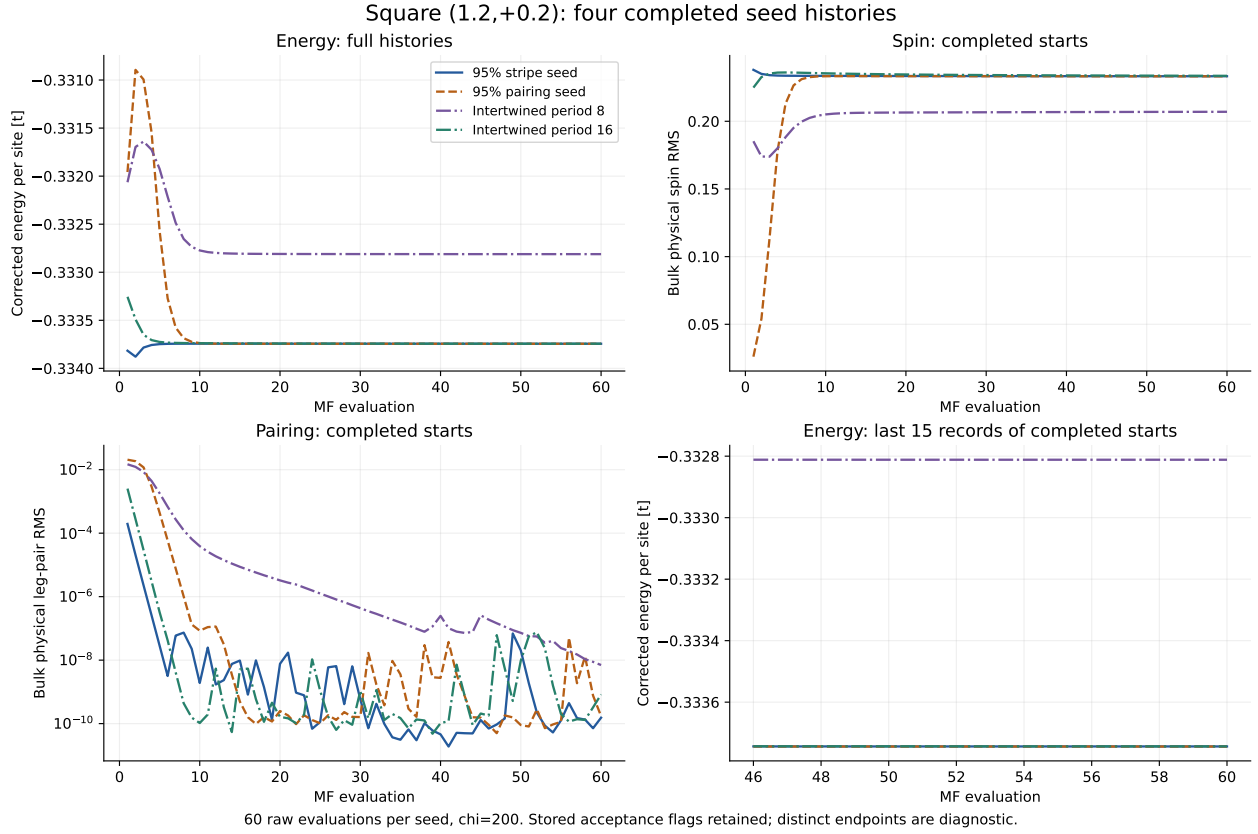


Figure 9: All four square (1.2, +0.2) histories. Full energy, spin RMS, logarithmic pairing RMS and late energy detail separate loss of pairing from persistence of different magnetic textures. Every start reaches the 60-evaluation cap without acceptance.

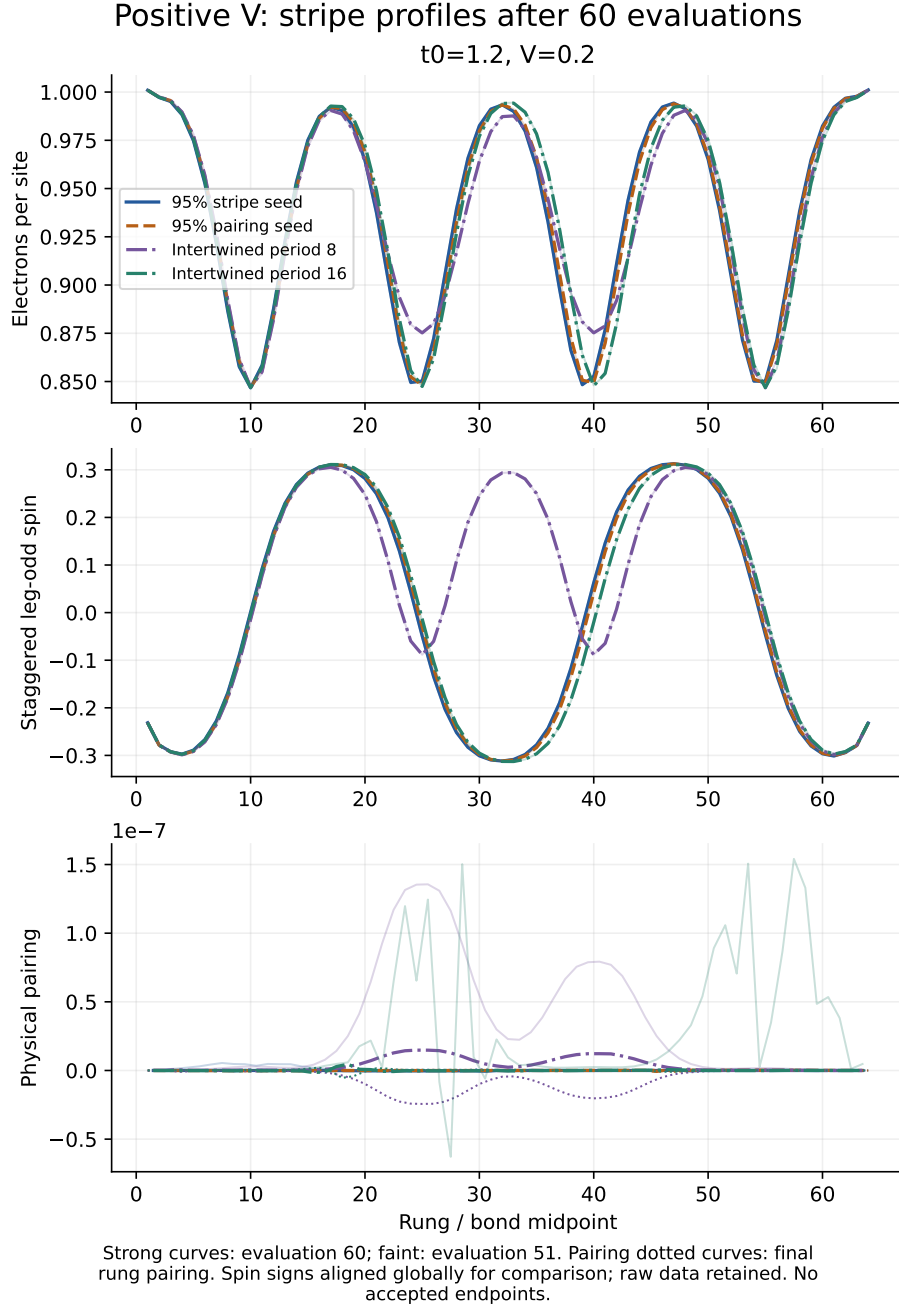


Figure 10: Square positive- V charge, staggered leg-odd spin and pairing profiles. Strong curves are evaluation 60, faint curves evaluation 51; dotted pairing curves are rung amplitudes. Only a global spin reversal aligns seeds. The period-eight start leaves two additional spin nodes. Pairing uses a 10^{-7} scale and is negligible in every endpoint; its residual spatial structure does not establish coexistence.

3.12 Trellis: spatial-cell dependence and alternating stripe relaxation

All four trellis comparisons now complete 60 raw cell sweeps at $(t_0/t, V/t) = (1, 0)$, $\tau_0/t = \tau_1/t = 0.1$, density $15/16$ and $\chi = 200$, with $L = 64$ per spatial ladder. Two seeds were tested in the skew one-ladder cell and two in the rectangular A/B cell. All four remain unaccepted maximum-iteration endpoints. The contrast in Fig. 11 is between spatial ansatzes: both one-ladder starts are pairing dominated, while both two-ladder starts develop strong, essentially unpaired stripes. The first one-ladder result must therefore be qualified by this additional cell dependence (Evidence 15).

In the one-ladder cell, the stripe seed loses spin from 0.0627 to 1.215×10^{-5} while leg-pair RMS rises from 0.00965 to 0.01796892 . The paired seed ends at spin RMS 1.368×10^{-5} and leg-pair RMS 0.01796878 . Both have mean leg/rung amplitudes near $+0.01796/ - 0.03292$: d-wave-like internal signs, not a spatial pair-density wave. Their energies, -0.518820336467 and -0.518820322856 t/site , differ by 1.361×10^{-8} t/site . Maximum seed-to-seed full-profile differences are 6.27×10^{-6} in charge and 5.68×10^{-7} in leg pairing. This supports a common paired basin, although final-ten energy ranges near 2.84×10^{-7} t/site and inner-DMRG/profile failures prevent acceptance. Small spin fluctuations persist near 10^{-5} .

Figure 12 contrasts the one-ladder paired profiles with the two-ladder stripes. Central charge standard deviation is about 0.00157 for one ladder, versus 0.0467 – 0.0470 for two. End-sensitive charge structure in the former does not establish bulk intertwining. The latter retains strong central modulation, spin RMS 0.22823 – 0.22876 on both spatial ladders, and leg-pair RMS no larger than 1.26×10^{-8} . Charge/spin bins $m = 4/30$ correspond to nominal periods $16/32$. Charge minima and spin nodes shift between A and B ; the latter is plotted at its physical offset of $-1/2$ rung. Profiles are read directly from stored correlations: trellis Hartree fields also contain normal-bond terms and cannot be inverted with the square/cubic density formula.

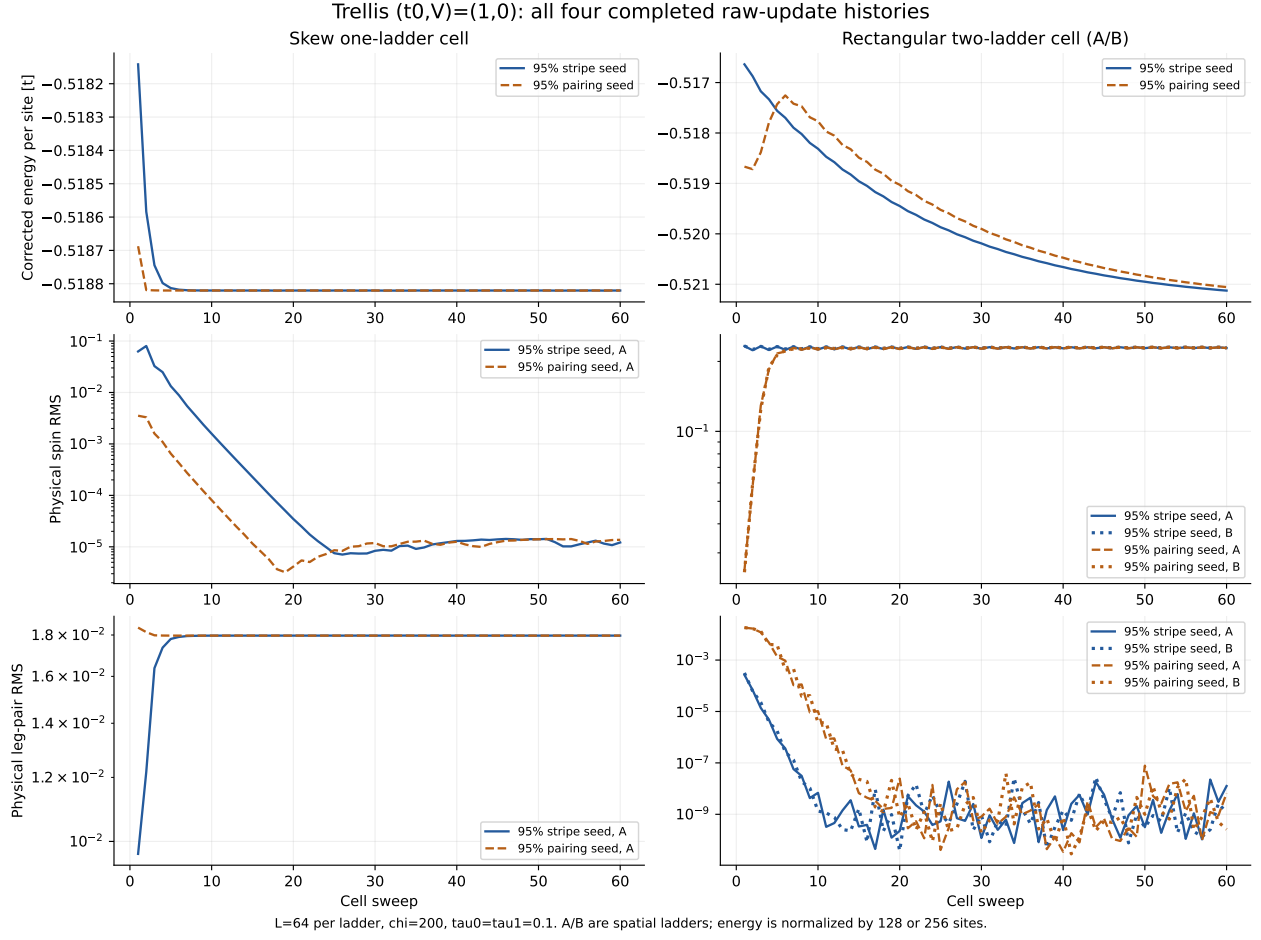


Figure 11: Complete trellis histories for both seeds in each spatial cell. Energy uses 128 sites for the skew one-ladder cell and 256 for the rectangular two-ladder cell. Spin and anomalous pairing use physical correlations and the same RMS windows as the square cuts. A/B are spatial ladders, not successive iterations or temporal orbit phases. Vertical scales differ between cells.

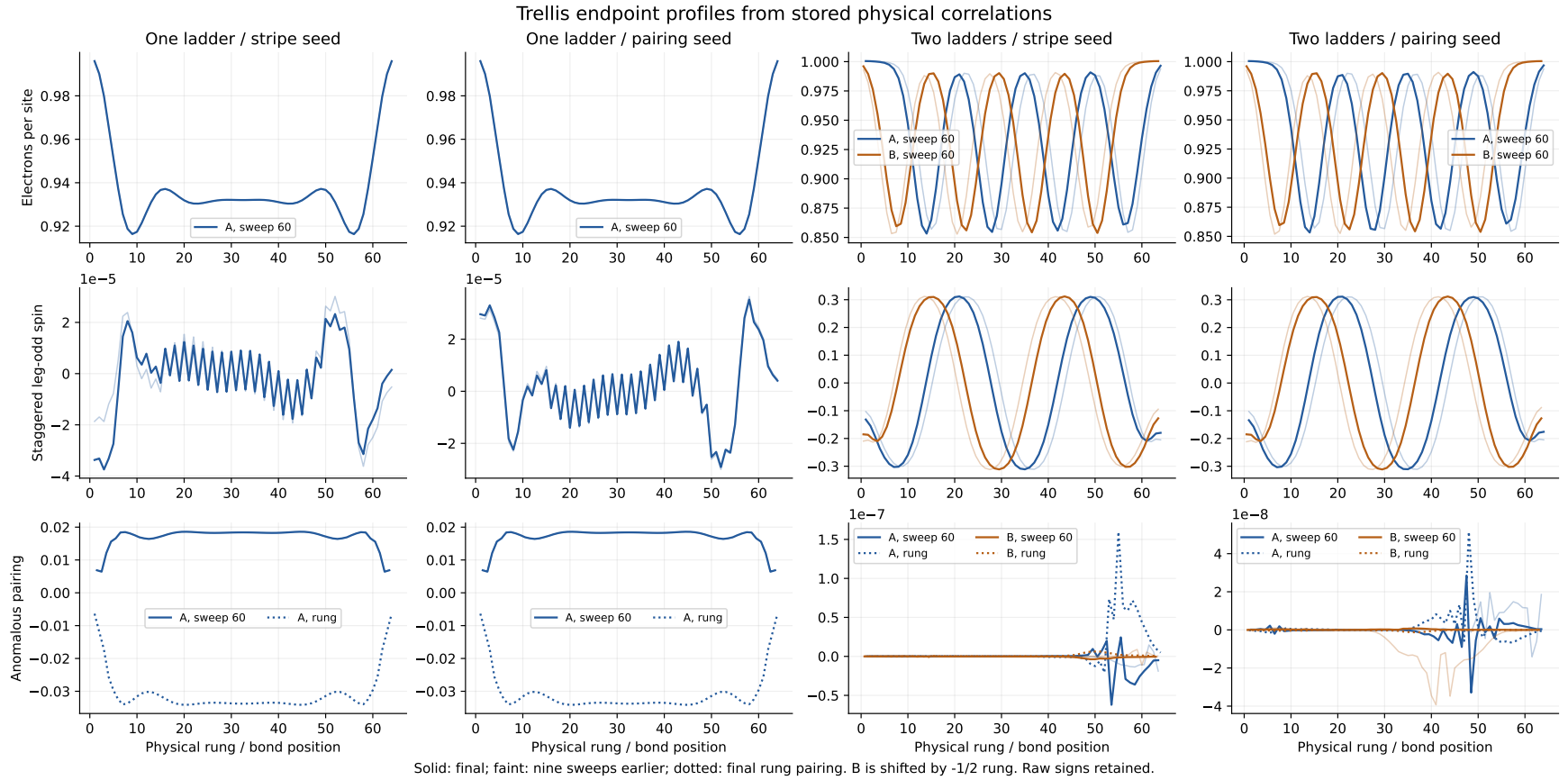


Figure 12: Trellis profiles for all four starts and six spatial ladders. Solid curves are the final sweep, faint curves nine sweeps earlier, and dotted curves final rung pairing. Raw signs are retained. One-ladder spin and two-ladder pairing panels use tiny scales; neither establishes an additional ordered component. The two-ladder spatial displacement changes with sweep parity, as resolved in Fig. 13.

Two-ladder nonacceptance is substantial, not marginal. Final global relative residuals are 0.1845–0.2085 against 10^{-4} allowed. Both final inner-DMRG windows pass, but the outer MF map and its spatial windows do not. Nearly constant spin RMS conceals the alternating wall displacement in Fig. 13. Successive full-cell measured-field increments have cosine -0.999962 and -0.999943 for the stripe and pairing starts, with norm ratios about 0.975. Maximum field changes are 0.02345/0.02620 over one sweep but 0.000754/0.000910 over two. The iteration nearly reverses its previous displacement while its amplitude slowly decreases. Two-sweep drift is still resolved. This is consistent with a damped alternating numerical mode, not an accepted period-two orbit or physical time-dependent order. Modest linear damping is a possible future diagnostic of that negative mode; no mixing change is made here.

The two-ladder stripe- and pairing-seeded endpoint energies are -0.521125816792 and $-0.521055560230 t/\text{site}$. Their difference, $7.026 \times 10^{-5} t/\text{site}$, is smaller than their summed final-ten ranges of $3.445 \times 10^{-4} t/\text{site}$. To resolve the finite-cell energy comparison, a September 20 audit places two identical copies of each saved paired ladder state in the rectangular cell and recomputes all transverse contractions there. Their target-corrected energies are -0.518817872689 and $-0.518817859078 t/\text{site}$, only $2.46378 \times 10^{-6} t/\text{site}$ above the original paired values. Both striped trials thus remain 0.002238–0.002308 t/site lower under the same finite-cell functional. Density corrections are below $8.33 \times 10^{-6} t/\text{site}$ and do not determine the ordering. The open-end embedding difference is consequently much too small to explain the observed energy advantage for these specified trials. Self-consistency is not needed to compare their expectation values. The large residuals still prevent identifying optimized stationary branches or a unique global phase minimum. Both seeds agree on the qualitative order within each cell, and a lower-energy striped trial has now been established in a common cell.

A remaining physical restriction is the transverse stripe period. After correcting the half-rung origin of B , the central charge harmonic at $q_x = \pi/8$ has a relative B -minus- A phase near 126–131 degrees. Repeating A/B alternates this phase advance and does not establish uniformly tilted, diagonal walls across many ladders. For an ideal charge profile $\delta n_m(x) = \rho \cos(q_x x + m\theta)$, a rectangular n -ladder cell requires $n\theta = 2\pi p$. Equivalently, for a rigid shift of d rungs per ladder, nd must close the longitudinal charge period and the full spin texture. Nominal charge/spin periods 16/32 with $d = 2$ would, for example, require sixteen ladders to repeat both without an additional spin transformation. This is an illustrative closure condition, not a measured tilt or proposed run size. The actual cell translation, trellis shear and leg basis must enter any such construction. Larger transverse cells or suitable translated repetitions could test oblique stripes; the current two-cell comparison does not determine their optimum.

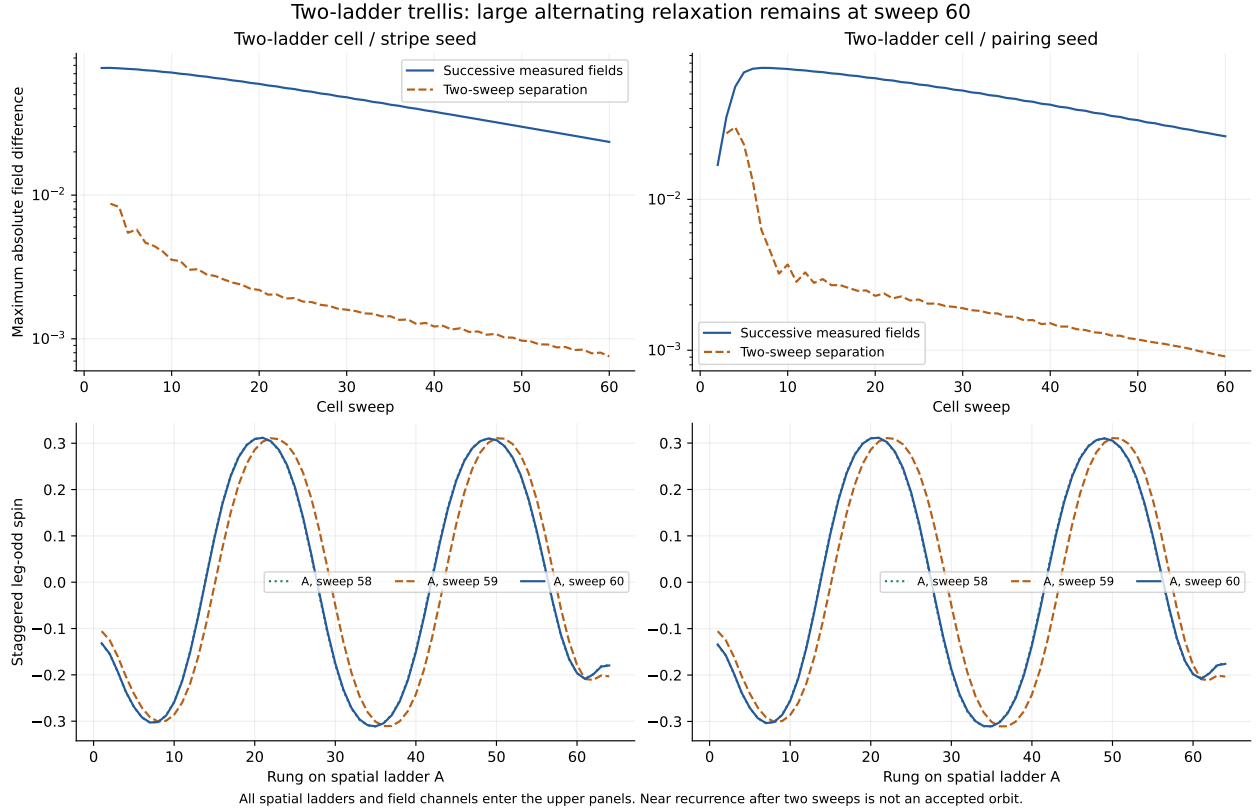


Figure 13: Two-ladder relaxation. Upper panels compare measured fields separated by one and two sweeps, taking the maximum over all channels and both spatial ladders. Lower panels resolve staggered spin on A at sweeps 58, 59 and 60. Even sweeps nearly overlap while the odd sweep has displaced nodes. Large one-step changes and continuing two-step drift preclude stationarity.

3.13 Scope of the emerging physical interpretation

The isolated-ladder and coupled-array observations separate pair formation from ordered-state selection. A spin-gapped ladder with slowly decaying pair correlations can still feed a static stripe state once transverse normal fields are allowed to develop. At the attractive anchor, both tested starts instead approach the same pairing-dominated solution. The current evidence therefore supports a model-specific competition between collective orders within a system that already possesses local pairing tendencies.

This result shares an ingredient with stripe-centered pairing mechanisms: correlated low-dimensional subsystems can support pairing before collective coherence is established. At the level of the current static textures, however, the anchor comparison resembles competition between stripe order and superconducting order. It neither establishes that fluctuating stripes are necessary for pairing nor excludes their possible beneficial role. The approximation treats quantum fluctuations within the ladder, but does not resolve the full collective stripe and phase fluctuations of a plane.

The most discriminating open physical question is whether the eventual stripe state retains strong pair-pair correlations after its anomalous expectation value has vanished. Strong residual pair correlations would support a distinction between local pairing and interladder coherence. Suppression of those correlations would instead indicate more direct competition with pairing. Resolving this requires four-fermion measurements on the stripe MPS; anomalous-amplitude histories alone do not answer it. These measurements are now enabled for future accepted and maximum-iteration endpoints, and a retrospective pass over the latest 56 branches (58 spatial MPSs) is prepared. Measuring an unaccepted endpoint is useful if its residuals and status accompany the correlations. No new pair-pair data from that pass enter the present analysis.

Attraction changes more than one ingredient of the coupled problem. The prefactor $g = t_{\perp}^2/|E_p|$ decreases when $|E_p|$ increases at fixed $t_{\perp}$, even while the ladder's pair response can become stronger. This motivates separating the change in the bare ladder from the change in the coupling kernel across a V scan. Recent MPS+MF work on a different, mixed-dimensional t - J model likewise emphasizes the properties of the one-dimensional building blocks and the requirements for an ordered array [48]. That work is retained as an August 2026 preprint, not a result for the present Hamiltonian.

3.14 Controls needed before stronger claims

The pair-breaking expansion needs a careful energy-scale comparison. The backbone gives $t_{\perp}/|E_p| \simeq 0.683$ and $t_{\perp}/\Delta_s \simeq 0.652$ at $t_{\perp}/t = 0.1$; these ratios are not parametrically small. The quantity

$$\Delta_{c,2} = \frac{E(N+2) + E(N-2) - 2E(N)}{2} \quad (13)$$

instead measures even-particle charge curvature. A gapless charge mode can coexist with a finite spin or pair-breaking scale, so $t_{\perp}/\Delta_{c,2} > 1$ does not by itself invalidate the projection. This corrects the stronger interpretation in the earliest bare-ladder report (Evidence 5). The registry and backbone also use slightly different binding energies, differing by about 0.0188% in the inferred coupling; quantitative response/SCF comparisons should declare a common kernel.

The remaining high-value controls are:

1. Complete self-consistency at the representative stripe point and compare matched canonical

energies only where the states and provenance support it. A converged fixed point and a global minimum remain different claims.

2. Test selected lengths, bond dimensions, and stripe wavelengths. The $L = 96$ and 128 seeds are prepared, but no result from them is used here. Their planned fixed- $L = 64$ binding denominator isolates one finite-size question; it is not a new measurement of the larger ladder's binding ([Evidence 16](#)).
3. Check sensitivity to the mean-field range cutoff, currently $r_{\text{range}} = 4$ in the representative comparisons, and to weaker transverse hopping. These are model/approximation controls, separate from inner-DMRG convergence.
4. Complete amplitude and basis checks for the response pilot and measure response about a self-consistent normal reference at fixed density.
5. Separate connected fluctuations from disconnected order contributions. Subtracting $\langle O_i \rangle \langle O_j \rangle$ removes the elastic mean-field order contribution; both parts are needed to interpret structure factors. Compare pair correlations across bond classes and bulk reference positions.

The completed square and unfrustrated-cubic grids now supply contrasting finite-run phase patterns. The finer cuts identify competing textures but do not resolve the order of the transition. The final September 19 review contains 38 completed 60-step run histories, or 2280 cell updates and 2400 individual ladder solves, all unaccepted; the earlier coarse square grid is a separate data set. No universal effect of V , certified ground-state phase diagram, transition temperature, phase stiffness, pair-density-wave state, or dynamical magnetic spectrum is claimed. The present contribution is a developing account of how local pairing tendencies and competing static orders relate in a specified coupled-ladder construction.

A Local evidence and drafting notes

A.1 Sources for project-specific statements

The following sources are repository documents and analysis artifacts, not external publications. Paths are relative to `ladder_mps_mft/docs`. The linked reports retain the underlying histories, definitions, and provenance checks. Their dated conclusions must be read together: later history analyses supersede the early suggestion of a unique paired $V = 0$ basin.

1. **Model, fields, and energy.** [Living methods notes](#), [variational functional](#), and [architecture](#). `METHODS_NOTES.tex`; `VARIATIONAL_FUNCTIONAL.md`; `ARCHITECTURE.md`.
2. **Literature and equation-level geometry correspondence.** [Literature review](#) and [source notes](#), including the September 8 comparison with the 2023 benchmark. `literature/literature_review.tex`; `literature/SOURCE_NOTES.md`.
3. **Isolated-ladder backbone and covariance survey, September 2.** [Stage 1 report](#). `reports/bare_stage1_t014_v0_20260902/REPORT.md`.
4. **Projected response discovery, September 3.** [Stage 2 report](#), with eigenvalues and reference-field summaries. `reports/bare_stage2_t014_v0_20260903/`.

5. **Cross-project scientific review, September 4.** [Review](#). Source for the response limitations, charge-gap correction, and proposed approximation controls; its campaign statuses are historical. `reports/systematic_review_20260904/REVIEW.md`.
6. **Bond-dimension comparison, September 5–6.** [Analysis](#), including the explicitly qualified September 6 energetic interpretation. `reports/chi400_comparison_20260905/ANALYSIS.md`.
7. **History-based revision of basin interpretation, September 8.** [Basin assessment](#). `reports/square_grid_20260908/BASIN_ASSESSMENT.md`.
8. **Square coverage and the flagged $(1.0, -0.4)$ endpoint, September 8.** [Grid description](#) and [divergence analysis](#). `reports/square_grid_20260908/`.
9. **Attractive two-family anchors, September 10.** [Analysis](#); complementary retrospective gate qualification in the [remainder contract](#). `reports/two_basin_vm04_20260910/README.md`.
10. **Zero-attraction anchors and slow texture relaxation, September 12–13.** [Analysis and follow-up](#). Figure 1 is reused without modifying the underlying data. `reports/two_basin_v000_20260912/README.md`.
11. **Combined synchronized campaign review, updated September 19.** [Combined report](#) and [complete cubic grid](#). Source inventories, physical profiles, gate checks and accounting coverage are retained with the detailed reports. The side-by-side figures use [matched physical histories](#) and their [source inventory](#).
12. **Complete coarse square grid and continuations.** [Updated square report](#), including both $(1.4, 0)$ continuations and the historical parent records.
13. **Finer square cuts and full energy curves, September 18.** [Cut report](#), [endpoint energy table](#), and [energy audit](#). Twenty endpoint sources include the twelve fine runs and eight coarse anchors. The [order-parameter audit](#) uses the same sources and consistent physical RMS windows.
14. **Square repulsive-interaction test, completed September 19.** [Four-seed comparison](#); all four spatial endpoints, with spin-node and energy diagnostics.
15. **All four trellis endpoints and alternating relaxation, September 19.** [Trellis report](#) and [spatial-cell method contract](#).
16. **Prepared finite-size seeds, September 6.** [Seed description](#). Preparation evidence only; no larger-length simulation result is inferred. `reports/finite_size_seeds_20260906/README.md`.

A.2 Reference provenance and manuscript development

The draft reuses stable keys from `../literature/references.bib`. The initial five additional sources in `additional_references.bib` are Tranquada et al. (2004), Vojta and Ulbricht (2004), Li et al. (2007), Berg et al. (2007), and Missiaen et al. (2025). Their primary arXiv and publisher records were checked on 15 September 2026. These additions support the discussion of magnetic spectra, planar coherence, and the stripe–pseudogap connection. Eleven further sources, checked

on 18 September 2026, support the ladder-material, doping, optical-parameter, charge-order, NMR, neutron, and RIXS discussion in Section 2.4; reading depths and primary links are recorded in `SOURCE_NOTES.md`. The existing review’s reading-depth limits continue to apply to inherited references; this draft is not an exhaustive literature or novelty assessment.

The introduction can be shortened for a target journal after the principal result and convergence controls are settled. The current results collection should retain its date and evidence qualifications until replaced by converged comparisons. A future manuscript should decide whether its central claim is interaction-driven order selection, geometry dependence, or the relation between a dressed normal-state response and the nonlinear ordered solutions. Those are candidate emphases, not additional results.

References

- [1] J. Hubbard. Electron correlations in narrow energy bands. *Proc. R. Soc. Lond. A*, 276(1365): 238–257, 1963. doi: 10.1098/rspa.1963.0204. URL <https://doi.org/10.1098/rspa.1963.0204>.
- [2] P. W. Anderson. The Resonating Valence Bond State in La_2CuO_4 and Superconductivity. *Science*, 235(4793):1196–1198, 1987. doi: 10.1126/science.235.4793.1196. URL <https://doi.org/10.1126/science.235.4793.1196>.
- [3] Patrick A. Lee, Naoto Nagaosa, and Xiao-Gang Wen. Doping a Mott insulator: Physics of high-temperature superconductivity. *Rev. Mod. Phys.*, 78(1):17–85, 2006. doi: 10.1103/RevModPhys.78.17. URL <https://doi.org/10.1103/RevModPhys.78.17>.
- [4] Daniel P. Arovas, Erez Berg, Steven A. Kivelson, and Srinivas Raghu. The Hubbard Model. *Annu. Rev. Condens. Matter Phys.*, 13(1):239–274, 2022. doi: 10.1146/annurev-conmatphys-031620-102024. URL <https://doi.org/10.1146/annurev-conmatphys-031620-102024>.
- [5] J. M. Tranquada, B. J. Sternlieb, J. D. Axe, Y. Nakamura, and S. Uchida. Evidence for stripe correlations of spins and holes in copper oxide superconductors. *Nature*, 375(6532):561–563, 1995. doi: 10.1038/375561a0. URL <https://doi.org/10.1038/375561a0>.
- [6] J. M. Tranquada, H. Woo, T. G. Perring, H. Goka, G. D. Gu, G. Xu, M. Fujita, and K. Yamada. Quantum magnetic excitations from stripes in copper oxide superconductors. *Nature*, 429:534–538, 2004. doi: 10.1038/nature02574. URL <https://doi.org/10.1038/nature02574>.
- [7] Q. Li, M. Hücker, G. D. Gu, A. M. Tsvelik, and J. M. Tranquada. Two-Dimensional Superconducting Fluctuations in Stripe-Ordered $\text{La}_{1.875}\text{Ba}_{0.125}\text{CuO}_4$. *Phys. Rev. Lett.*, 99:067001, 2007. doi: 10.1103/PhysRevLett.99.067001. URL <https://doi.org/10.1103/PhysRevLett.99.067001>.
- [8] Erez Berg, Eduardo Fradkin, Eun-Ah Kim, Steven A. Kivelson, Vadim Oganesyan, John M. Tranquada, and Shoucheng Zhang. Dynamical Layer Decoupling in a Stripe-Ordered High- T_c Superconductor. *Phys. Rev. Lett.*, 99:127003, 2007. doi: 10.1103/PhysRevLett.99.127003. URL <https://doi.org/10.1103/PhysRevLett.99.127003>.
- [9] Eduardo Fradkin, Steven A. Kivelson, and John M. Tranquada. Colloquium: Theory of intertwined orders in high temperature superconductors. *Rev. Mod. Phys.*, 87(2):457–482, 2015. doi: 10.1103/RevModPhys.87.457. URL <https://doi.org/10.1103/RevModPhys.87.457>.
- [10] Daniel F. Agterberg, J.C. Séamus Davis, Stephen D. Edkins, Eduardo Fradkin, Dale J. Van Harlingen, Steven A. Kivelson, Patrick A. Lee, Leo Radzihovsky, John M. Tranquada, and Yuxuan Wang. The Physics of Pair-Density Waves: Cuprate Superconductors and Beyond. *Annu. Rev. Condens. Matter Phys.*, 11(1):231–270, 2020. doi: 10.1146/annurev-conmatphys-031119-050711. URL <https://doi.org/10.1146/annurev-conmatphys-031119-050711>.
- [11] Bo-Xiao Zheng, Chia-Min Chung, Philippe Corboz, Georg Ehlers, Ming-Pu Qin, Reinhard M. Noack, Hao Shi, Steven R. White, Shiwei Zhang, and Garnet Kin-Lic Chan. Stripe order in the underdoped region of the two-dimensional Hubbard model. *Science*, 358(6367):1155–1160, 2017. doi: 10.1126/science.aam7127. URL <https://doi.org/10.1126/science.aam7127>.
- [12] Mingpu Qin, Chia-Min Chung, Hao Shi, Ettore Vitali, Claudius Hubig, Ulrich Schollwöck, Steven R. White, Shiwei Zhang, and Simons Collaboration on the Many-Electron Problem. Absence of Superconductivity in the Pure Two-Dimensional Hubbard Model. *Phys. Rev. X*, 10(3):031016, 2020. doi: 10.1103/PhysRevX.10.031016. URL <https://doi.org/10.1103/PhysRevX.10.031016>.
- [13] Hong-Chen Jiang and Thomas P. Devereaux. Superconductivity in the doped Hubbard model

- and its interplay with next-nearest hopping t' . *Science*, 365(6460):1424–1428, 2019. doi: 10.1126/science.aal5304. URL <https://doi.org/10.1126/science.aal5304>.
- [14] Hao Xu, Chia-Min Chung, Mingpu Qin, Ulrich Schollwöck, Steven R. White, and Shiwei Zhang. Coexistence of superconductivity with partially filled stripes in the Hubbard model. *Science*, 384(6696):eadh7691, 2024. doi: 10.1126/science.adh7691. URL <https://doi.org/10.1126/science.adh7691>.
 - [15] Hong-Chen Jiang and Steven A. Kivelson. Stripe order enhanced superconductivity in the Hubbard model. *Proc. Natl. Acad. Sci. U.S.A.*, 119(1):e2109406119, 2022. doi: 10.1073/pnas.2109406119. URL <https://doi.org/10.1073/pnas.2109406119>.
 - [16] E. Dagotto, J. Riera, and D. Scalapino. Superconductivity in ladders and coupled planes. *Phys. Rev. B*, 45(10):5744–5747, 1992. doi: 10.1103/PhysRevB.45.5744. URL <https://doi.org/10.1103/PhysRevB.45.5744>.
 - [17] R. M. Noack, S. R. White, and D. J. Scalapino. Correlations in a Two-Chain Hubbard Model. *Phys. Rev. Lett.*, 73(6):882–885, 1994. doi: 10.1103/PhysRevLett.73.882. URL <https://doi.org/10.1103/PhysRevLett.73.882>.
 - [18] Elbio Dagotto and T. M. Rice. Surprises on the Way from One- to Two-Dimensional Quantum Magnets: The Ladder Materials. *Science*, 271(5249):618–623, 1996. doi: 10.1126/science.271.5249.618. URL <https://doi.org/10.1126/science.271.5249.618>.
 - [19] A. Luther and V. J. Emery. Backward Scattering in the One-Dimensional Electron Gas. *Phys. Rev. Lett.*, 33(10):589–592, 1974. doi: 10.1103/PhysRevLett.33.589. URL <https://doi.org/10.1103/PhysRevLett.33.589>.
 - [20] Leon Balents and Matthew P. A. Fisher. Weak-coupling phase diagram of the two-chain Hubbard model. *Phys. Rev. B*, 53(18):12133–12141, 1996. doi: 10.1103/PhysRevB.53.12133. URL <https://doi.org/10.1103/PhysRevB.53.12133>.
 - [21] Michele Dolfi, Bela Bauer, Sebastian Keller, and Matthias Troyer. Pair correlations in doped Hubbard ladders. *Phys. Rev. B*, 92(19):195139, 2015. doi: 10.1103/PhysRevB.92.195139. URL <https://doi.org/10.1103/PhysRevB.92.195139>.
 - [22] Yang Shen, Guang-Ming Zhang, and Mingpu Qin. Reexamining doped two-legged Hubbard ladders. *Phys. Rev. B*, 108(16):165113, 2023. doi: 10.1103/PhysRevB.108.165113. URL <https://doi.org/10.1103/PhysRevB.108.165113>.
 - [23] Masatomo Uehara, Takashi Nagata, Jun Akimitsu, Hiroki Takahashi, Nobuo Môri, and Kyoichi Kinoshita. Superconductivity in the Ladder Material $\text{Sr}_{0.4}\text{Ca}_{13.6}\text{Cu}_{24}\text{O}_{41.84}$. *J. Phys. Soc. Jpn.*, 65(9):2764–2767, 1996. doi: 10.1143/JPSJ.65.2764. URL <https://doi.org/10.1143/JPSJ.65.2764>.
 - [24] T. Nagata, M. Uehara, J. Goto, J. Akimitsu, N. Motoyama, H. Eisaki, S. Uchida, H. Takahashi, T. Nakanishi, and N. Môri. Pressure-Induced Dimensional Crossover and Superconductivity in the Hole-Doped Two-Leg Ladder Compound $\text{Sr}_{14-x}\text{Ca}_x\text{Cu}_{24}\text{O}_{41}$. *Phys. Rev. Lett.*, 81(5):1090–1093, 1998. doi: 10.1103/PhysRevLett.81.1090. URL <https://doi.org/10.1103/PhysRevLett.81.1090>.
 - [25] P. Abbamonte, G. Blumberg, A. Rusydi, A. Gozar, P. G. Evans, T. Siegrist, L. Venema, H. Eisaki, E. D. Isaacs, and G. A. Sawatzky. Crystallization of charge holes in the spin ladder of $\text{Sr}_{14}\text{Cu}_{24}\text{O}_{41}$. *Nature*, 431(7012):1078–1081, 2004. doi: 10.1038/nature02925. URL <https://doi.org/10.1038/nature02925>.
 - [26] A. Rusydi, P. Abbamonte, H. Eisaki, Y. Fujimaki, G. Blumberg, S. Uchida, and G. A. Sawatzky. Quantum Melting of the Hole Crystal in the Spin Ladder of $\text{Sr}_{14-x}\text{Ca}_x\text{Cu}_{24}\text{O}_{41}$. *Phys. Rev.*

- Lett.*, 97:016403, 2006. doi: 10.1103/PhysRevLett.97.016403. URL <https://doi.org/10.1103/PhysRevLett.97.016403>.
- [27] V. J. Emery, S. A. Kivelson, and O. Zachar. Spin-gap proximity effect mechanism of high-temperature superconductivity. *Phys. Rev. B*, 56(10):6120–6147, 1997. doi: 10.1103/PhysRevB.56.6120. URL <https://doi.org/10.1103/PhysRevB.56.6120>.
 - [28] E. Arrigoni, E. Fradkin, and S. A. Kivelson. Mechanism of high-temperature superconductivity in a striped Hubbard model. *Phys. Rev. B*, 69(21):214519, 2004. doi: 10.1103/PhysRevB.69.214519. URL <https://doi.org/10.1103/PhysRevB.69.214519>.
 - [29] Gunnar Bollmark, Nicolas Laflorencie, and Adrian Kantian. Dimensional crossover and phase transitions in coupled chains: Density matrix renormalization group results. *Phys. Rev. B*, 102(19):195145, 2020. doi: 10.1103/PhysRevB.102.195145. URL <https://doi.org/10.1103/PhysRevB.102.195145>.
 - [30] Gunnar Bollmark, Thomas Köhler, Lorenzo Pizzino, Yiqi Yang, Johannes S. Hofmann, Hao Shi, Shiwei Zhang, Thierry Giamarchi, and Adrian Kantian. Solving 2D and 3D Lattice Models of Correlated Fermions—Combining Matrix Product States with Mean-Field Theory. *Phys. Rev. X*, 13(1):011039, 2023. doi: 10.1103/PhysRevX.13.011039. URL <https://doi.org/10.1103/PhysRevX.13.011039>.
 - [31] Gunnar Bollmark, Thomas Köhler, and Adrian Kantian. Resolving competition of charge density wave and superconducting phases using the matrix product state plus mean field algorithm. *Phys. Rev. B*, 111(12):125141, 2025. doi: 10.1103/PhysRevB.111.125141. URL <https://doi.org/10.1103/PhysRevB.111.125141>.
 - [32] Zongsheng Zhou, Weinan Ye, Hong-Gang Luo, Jize Zhao, and Jun Chang. Robust superconducting correlation against intersite interactions in the extended two-leg Hubbard ladder. *Phys. Rev. B*, 108(19):195136, 2023. doi: 10.1103/PhysRevB.108.195136. URL <https://doi.org/10.1103/PhysRevB.108.195136>.
 - [33] Cheng Peng, Yao Wang, Jiajia Wen, Young S. Lee, Thomas P. Devereaux, and Hong-Chen Jiang. Enhanced superconductivity by near-neighbor attraction in the doped extended Hubbard model. *Phys. Rev. B*, 107(20):L201102, 2023. doi: 10.1103/PhysRevB.107.L201102. URL <https://doi.org/10.1103/PhysRevB.107.L201102>.
 - [34] Hari Padma, Jinu Thomas, Sophia F. R. TenHuisen, Wei He, Ziqiang Guan, Jiemin Li, Byungjune Lee, Yu Wang, Seng Huat Lee, Zhiqiang Mao, Hoyoung Jang, Valentina Bisogni, Jonathan Pelliciari, Mark P. M. Dean, Steven Johnston, and Matteo Mitrano. Beyond-Hubbard Pairing in a Cuprate Ladder. *Phys. Rev. X*, 15(2):021049, 2025. doi: 10.1103/PhysRevX.15.021049. URL <https://doi.org/10.1103/PhysRevX.15.021049>.
 - [35] A. Missiaen, H. Mayaffre, S. Krämer, D. Zhao, Y. B. Zhou, T. Wu, X. H. Chen, S. Pyon, T. Takayama, H. Takagi, D. LeBoeuf, and M.-H. Julien. Spin-Stripe Order Tied to the Pseudogap Phase in $\text{La}_{1.8-x}\text{Eu}_{0.2}\text{Sr}_x\text{CuO}_4$. *Phys. Rev. X*, 15:021010, 2025. doi: 10.1103/PhysRevX.15.021010. URL <https://doi.org/10.1103/PhysRevX.15.021010>.
 - [36] Matthias Vojta and Tobias Ulbricht. Magnetic Excitations in a Bond-Centered Stripe Phase: Spin Waves Far from the Semiclassical Limit. *Phys. Rev. Lett.*, 93:127002, 2004. doi: 10.1103/PhysRevLett.93.127002. URL <https://doi.org/10.1103/PhysRevLett.93.127002>.
 - [37] M. Azuma, Z. Hiroi, M. Takano, K. Ishida, and Y. Kitaoka. Observation of a Spin Gap in SrCu_2O_3 Comprising Spin- $\frac{1}{2}$ Quasi-1D Two-Leg Ladders. *Phys. Rev. Lett.*, 73:3463–3466, 1994. doi: 10.1103/PhysRevLett.73.3463. URL <https://doi.org/10.1103/PhysRevLett.73.3463>.
 - [38] Z. Hiroi and M. Takano. Absence of superconductivity in the doped antiferromagnetic spin-ladder compound $(\text{La,Sr})\text{CuO}_{2.5}$. *Nature*, 377:41–43, 1995. doi: 10.1038/377041a0. URL

- <https://doi.org/10.1038/377041a0>.
- [39] B. Normand and T. M. Rice. Electronic and magnetic structure of $\text{LaCuO}_{2.5}$. *Phys. Rev. B*, 54:7180–7188, 1996. doi: 10.1103/PhysRevB.54.7180. URL <https://doi.org/10.1103/PhysRevB.54.7180>.
 - [40] Z. V. Popović, M. J. Konstantinović, V. A. Ivanov, O. P. Khuong, R. Gajić, A. Vietkin, and V. V. Moshchalkov. Optical properties of the spin-ladder compound $\text{Sr}_{14}\text{Cu}_{24}\text{O}_{41}$. *Phys. Rev. B*, 62:4963–4972, 2000. doi: 10.1103/PhysRevB.62.4963. URL <https://doi.org/10.1103/PhysRevB.62.4963>.
 - [41] N. Fujiwara, Y. Fujimaki, S. Uchida, K. Matsubayashi, T. Matsumoto, and Y. Uwatoko. NMR and NQR study of pressure-induced superconductivity and the origin of critical-temperature enhancement in the spin-ladder cuprate $\text{Sr}_2\text{Ca}_{12}\text{Cu}_{24}\text{O}_{41}$. *Phys. Rev. B*, 80:100503, 2009. doi: 10.1103/PhysRevB.80.100503. URL <https://doi.org/10.1103/PhysRevB.80.100503>.
 - [42] S. Frank, A. Huber, U. Ammerahl, M. Hückler, and C. A. Kuntscher. Polarization-dependent infrared reflectivity study of $\text{Sr}_{2.5}\text{Ca}_{11.5}\text{Cu}_{24}\text{O}_{41}$ under pressure: Charge dynamics, charge distribution, and anisotropy. *Phys. Rev. B*, 90:224516, 2014. doi: 10.1103/PhysRevB.90.224516. URL <https://doi.org/10.1103/PhysRevB.90.224516>.
 - [43] T. Vuletić, B. Korin-Hamzić, S. Tomić, B. Gorshunov, P. Haas, T. Rößm, M. Dressel, J. Akimitsu, T. Sasaki, and T. Nagata. Suppression of the Charge-Density-Wave State in $\text{Sr}_{14}\text{Cu}_{24}\text{O}_{41}$ by Calcium Doping. *Phys. Rev. Lett.*, 90:257002, 2003. doi: 10.1103/PhysRevLett.90.257002. URL <https://doi.org/10.1103/PhysRevLett.90.257002>.
 - [44] S. Katano, T. Nagata, H. Fujino, J. Akimitsu, M. Nishi, and K. Kakurai. Pressure effects on spin-gap and antiferromagnetism in the hole-doped spin ladder system $\text{Sr}_{2.5}\text{Ca}_{11.5}\text{Cu}_{24}\text{O}_{41}$. *Physica B*, 259–261:1046–1047, 1999. doi: 10.1016/S0921-4526(98)00692-9. URL [https://doi.org/10.1016/S0921-4526\(98\)00692-9](https://doi.org/10.1016/S0921-4526(98)00692-9).
 - [45] Naoki Fujiwara, Nobuo Môri, Yoshiya Uwatoko, Takehiko Matsumoto, Naoki Motoyama, and Shinichi Uchida. Superconductivity of the $\text{Sr}_2\text{Ca}_{12}\text{Cu}_{24}\text{O}_{41}$ Spin-Ladder System: Are the Superconducting Pairing and the Spin-Gap Formation of the Same Origin? *Phys. Rev. Lett.*, 90:137001, 2003. doi: 10.1103/PhysRevLett.90.137001. URL <https://doi.org/10.1103/PhysRevLett.90.137001>.
 - [46] Y. Tseng, J. Thomas, W. Zhang, E. Paris, P. Puphal, R. Bag, G. Deng, T. C. Asmara, V. N. Strocov, S. Singh, E. Pomjakushina, U. Kumar, A. Nocera, H. M. Rønnow, S. Johnston, and T. Schmitt. Crossover of high-energy spin fluctuations from collective triplons to localized magnetic excitations in $\text{Sr}_{14-x}\text{Ca}_x\text{Cu}_{24}\text{O}_{41}$ ladders. *npj Quantum Mater.*, 7:92, 2022. doi: 10.1038/s41535-022-00502-1. URL <https://doi.org/10.1038/s41535-022-00502-1>.
 - [47] Steven R. White, Ian Affleck, and Douglas J. Scalapino. Friedel oscillations and charge density waves in chains and ladders. *Phys. Rev. B*, 65(16):165122, 2002. doi: 10.1103/PhysRevB.65.165122. URL <https://doi.org/10.1103/PhysRevB.65.165122>.
 - [48] Thomas Köhler and Adrian Kantian. Predicting critical temperature in quantum simulators for high- T_c superconductivity: the matrix product state plus mean field approach. arXiv preprint, 2026. URL <https://arxiv.org/abs/2608.18861>. Preprint version; publication status checked 7 September 2026.